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FACULTY OF SCIENCE
DEPARTMENT OF COMPUTER AND INFORMATION SCIENCE

Integrated Requisition Management System (Reqon):

A Case of Isuzu East Africa Ltd.

BY

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A project report submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of
Science in Computer Science

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DECLARATION AND APPROVAL

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I, the undersigned, declare that this project is my original work and that it has not been presented in any other university or institution for academic credit.

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DEDICATION

This project is dedicated to my family for their unwavering love, encouragement and support throughout my academic journey. Their belief in my abilities has been a constant source of motivation and strength.

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ABSTRACT

Requisition processes are central to institutional operations, supporting functions such as procurement, personnel management, maintenance services and IT asset management. However, many organizations continue to rely on fragmented manual or semi-digital systems such as emails and paper-based forms. These approaches introduce inefficiencies in managing requisition and approval processes due to manual workflows, disjointed communication channels, and limited visibility into requisition status often resulting in delayed approvals, duplicated requests, and poor traceability. The objective of this project is to design and develop a web-based Requisition Management System (Recon) that consolidates multiple institutional requisition processes into a centralized system. The primary objective of the system is to streamline end-to-end requisition workflows, automate approval processes, improve visibility across approval stages, and enhance accountability through audit trails and structured approval hierarchies. The system enables users to submit, review, approve, and track requisitions through role-based workflows. Key features include requisition submission and tracking, multi-level approval workflows, automated notifications, role-based access control, audit logging, dashboard analytics, user management, and the generation of Local Purchase Orders (LPOs) from fully approved requisitions, thereby integrating requisition processing with procurement execution. Management dashboards provide real-time insights into requisition trends, approval performance, and expenditure patterns, supporting data-driven decision-making. The system was developed using a structured system development methodology comprising requirements analysis, system design, implementation, and testing. The frontend was developed using HTML, CSS, and JavaScript, while PHP was used for server-side processing and business logic implementation. A MySQL relational database was used to store requisition records, approval histories, user information, and audit logs. Testing results demonstrate that the system successfully automates requisition and approval workflows, improves process transparency, enhances procurement efficiency through LPO generation, and provides valuable operational insights. This project addresses the need for organizations to modernize requisition and approval processes through a functional and scalable system while establishing a foundation for future enhancements such as intelligent decision support and scalable enterprise integration.

TABLE OF CONTENTS

DECLARATION AND APPROVAL.....	1
ACKNOWLEDGEMENT.....	2
DEDICATION.....	3
ABSTRACT.....	4
TABLE OF CONTENTS.....	5
LIST OF FIGURES.....	8
LIST OF TABLES.....	9
DEFINITION OF KEY TERMS.....	10
CHAPTER ONE: INTRODUCTION.....	1
1.1 Background and Motivation.....	1
1.2 Problem Statement.....	2
1.3 Aim of Research.....	2
1.4 Objectives of the Research.....	2
1.5 Justification of the Research.....	3
1.6 Scope.....	4
1.7 Report Organization.....	5
CHAPTER 2: REVIEW OF RELATED WORK.....	6
2.1 Chapter Introduction.....	6
2.2 History of the Research Topic.....	6
2.3 Review of Related Prototypes and Systems.....	7
2.3.1 Enterprise Resource Planning (ERP) Systems.....	7
2.3.2 Commercial Requisition and Procurement Platforms.....	8
2.3.3 Public Sector E-Requisition Systems in Kenya.....	9
2.3.4 Academic Prototype: Web-Based Material Requisition System.....	9
2.3.5 Intelligent and AI-Driven Requisition Systems.....	10
2.4 Emerging Trends and Patterns.....	11
2.5 Chapter Summary.....	11
CHAPTER 3: RESEARCH METHODOLOGY.....	12
3.1 Chapter Introduction.....	12
3.2 Methodology of Requirement Specification and Data Collection.....	12
3.2.1 Interviews.....	12
3.2.2 Observation.....	12
3.2.3 Document Review/Analysis.....	13
3.3 Methodology for System Analysis of the Current System.....	13
3.4 Methodology for System Design (Proposed System).....	13
3.5 Methodology for System Implementation.....	14
3.5.1 Tools and Technology.....	14
Front-end:.....	14
Back-end:.....	14

Development Environment.....	14
3.6 Methodology for System Testing.....	15
3.6.1 Testing Plan.....	15
3.6.2 Testing Techniques.....	15
3.7 Methodology for System Deployment.....	15
3.7.1 Deployment Strategy.....	15
3.7.2 Data Migration Plan.....	16
3.8 Chapter Summary.....	16
CHAPTER 4 : SYSTEM ANALYSIS.....	17
4.1 Chapter Introduction.....	17
4.2 Description of the Current System.....	17
4.3 Strengths of the Current System.....	18
4.4 Weaknesses of the Current System.....	18
4.5 Feasibility Study and Conclusion.....	19
4.5.1 Operational Feasibility.....	19
4.5.2 Technical Feasibility.....	19
4.5.3 Economic Feasibility.....	20
4.6 System Analysis.....	22
4.6.1 Structured System Analysis and Design (SSAD).....	22
4.6.2 Context Diagram.....	22
4.6.3 Level 1 Data Flow Diagram (DFD).....	23
4.6.4 System Users and Actors.....	24
4.6.5 Process Flowchart.....	25
4.7 Chapter Summary.....	27
CHAPTER 5: SYSTEM DESIGN.....	28
5.1 Chapter Introduction.....	28
5.2 Description of the Proposed System.....	28
5.3 Strengths of the Proposed System.....	29
5.4 Weaknesses of the Proposed System.....	29
5.5 Requirements Analysis.....	30
5.5.1 Functional Requirements.....	30
5.5.2 Non-Functional Requirements.....	31
5.6 Conceptual Architecture of the Proposed System.....	33
5.7 Process Design.....	34
5.7.1 Proposed System Context Diagram.....	34
5.7.2 Level 1 DFD.....	35
5.7.3 Flowcharts.....	35
5.7.4 Sequence Diagram.....	39
5.8 Entity - Relationship Diagram (ERD).....	40
5.9 Data Dictionary.....	40
5.9.1 Users Table.....	40

5.9.2 Departments Table.....	41
5.9.3 Roles Table.....	42
5.9.4 Requisitions Table.....	42
5.9.5 Requisition_Items Table.....	43
5.9.6 Item_Catalog Table.....	44
5.9.7 Approval_Levels.....	45
5.9.8 Approval_History Table.....	46
5.9.9 Notifications Table.....	47
5.9.10 Lpo_log table.....	48
5.9.11 Audit_Log Table.....	49
5.9 Proposed System Prototype.....	50
5.10 Chapter Summary.....	52
CHAPTER 6: SYSTEM IMPLEMENTATION AND TESTING.....	53
6.1 Chapter Introduction.....	53
6.2 System Implementation.....	53
6.3 System Screenshots.....	54
6.4 Testing Plan.....	56
6.4.1 Unit Testing.....	58
6.4.2 Integration Testing.....	58
6.4.3 System Testing.....	59
6.5 Evaluation Plan.....	60
6.5.1 Evaluation of Objective 1: Automate Requisition Creation and Submission.....	60
6.5.2 Evaluation of Objective 2: Automate Approval Routing and Workflow Management.....	61
6.5.3 Evaluation of Objective 3: Improve Visibility and Tracking of Requisitions.....	61
6.5.4 Evaluation of Objective 4: Improve Procurement Support through LPO Generation.....	62
6.5.5 Evaluation of Objective 5: Support Management Decision-Making.....	62
6.6 Chapter Summary.....	63
CHAPTER 7: CONCLUSIONS, FINDINGS AND RECOMMENDATIONS.....	64
7.1 Chapter Introduction.....	64
7.2 Conclusion.....	64
7.3 Key Findings.....	65
7.4 Challenges Encountered.....	67
7.5 Future Recommendations.....	67
7.6 Summary.....	68
Appendix A: Project Schedule (Gantt chart).....	69
Appendix B: Project Budget and Resources.....	70
Appendix C: Sample Interview Guide.....	71
Appendix D: Sample Code.....	72
Appendix E: Manual Requisition Form.....	74
REFERENCES.....	75

LIST OF FIGURES

Figure 2.1: Spendflo system screenshot.....	9
Figure 2.2: Material Requisition System prototype screenshot.....	10
Figure 4.1: Current System Context Diagram.....	23
Figure 4.2: Current System Level1 DFD.....	24
Figure 4.3: Current Process Flowchart - Requester.....	26
Figure 5.1: Proposed System Context Diagram.....	34
Figure 5.2: Proposed System Level 1 DFD.....	35
Figure 5.3: User Login Process Flowchart.....	36
Figure 5.4: Requisition Submission Process.....	37
Figure 5.5: Approval Decision Process.....	38
Figure 5.6: Sequence Diagram.....	39
Figure 5.7: Entity Relationship Diagram (ERD).....	40
Screen 1: Login Page Design.....	50
Screen 2: Dashboard Page Design.....	51
Screen 3: Digital Requisition Form.....	51
Screen 4: Requisition Details Page (Report).....	52
Figure 6.1: Login Interface.....	54
Figure 6.2 Dashboard screen.....	55
Figure 6.3: 3-Step Requisition Form.....	55
Figure 6.4: LPO Generated by the System after approvals.....	56

LIST OF TABLES

TABLE 4.1: ESTIMATED ANNUAL COST COMPARISON.....	21
TABLE 4.2: USER ROLES, INPUTS, AND OUTPUTS.....	24
TABLE 5.1: USERS TABLE.....	41
TABLE 5.2: DEPARTMENTS TABLE.....	41
TABLE 5.3: ROLES TABLE.....	42
TABLE 5.4: REQUISITIONS TABLE.....	42
TABLE 5.5: REQUISITION_ITEMS TABLE.....	43
TABLE 5.6: ITEM_CATALOG TABLE.....	44
TABLE 5.7: APPROVAL_LEVELS TABLE.....	45
TABLE 5.8: APPROVAL_HISTORY TABLE.....	46
TABLE 5.9: NOTIFICATIONS TABLE.....	47
TABLE 5.10: LPO_LOG TABLE.....	48
TABLE 5.11: AUDIT_LOG TABLE.....	49
TABLE 6.1: SAMPLE TEST CASES.....	59
TABLE 1: PROJECT SCHEDULE.....	69
TABLE 2: PROJECT BUDGET.....	70

DEFINITION OF KEY TERMS

DFD	Data Flow Diagram. Tool that shows how information moves through a system.
ERD	Entity Relationship Diagram. Tool used in database design.
ERP	Enterprise Resource Planning
GEP	Global eProcure Global leading company in AI-powered Procurement and Supply Chain
IT	Information Technology
LPO	Local Purchase Order. An official written document transmitted from a purchaser to a supplier, granting permission for the acquisition of particular commodities.
SAP	Systems, Applications and Products in Data Processing. A German software company and market leader in ERP software.
SSAD	Structured System Analysis and Design
SQL	Structured Query Language
Requisition	A formal request submitted by an employee or department seeking approval to obtain goods, services, personnel resources, or other organizational requirements. It serves as the initial step in the procurement or approval process.

CHAPTER ONE: INTRODUCTION

1.1 Background and Motivation

Information Technology (IT) has become a pivotal element in the operations of contemporary organizations, shaping how businesses conduct, manage and optimize their activities. The rapid advancement of IT infrastructure and applications has transformed organizational landscapes by enhancing communication, automating routine tasks, and supporting complex decision-making processes. This technological evolution has enabled organizations to achieve higher efficiency, agility, and competitive advantage in an increasingly global and dynamic market environment (Brynjolfsson & McAfee, 2017).

Across multiple sectors, IT's impacts have been profound and varied. In manufacturing, automation and enterprise resource planning (ERP) systems optimize production lines and supply chains, reducing waste and improving product quality (Kumar & Kumar, 2021). The healthcare sector has seen improvements in patient care and data management enhancing operational efficiency and treatment outcomes (Raghupathi & Raghupathi, 2020). Financial organizations utilize IT extensively for real-time transaction processing, risk management, and fraud detection, facilitating faster and more secure services (Chen et al., 2019). Moreover, the public sector has embraced IT for e-governance initiatives to improve transparency, service delivery, and citizen management (Heeks & Bannister, 2018).

Requisition management involves the initiation, approval, and tracking of requests within organizations, directly impacting supply chain efficiency and resource utilization. Observations from organizational environments reveal that requisition activities are often handled using semi-manual approaches such as emails, spreadsheets, and paper-based forms, creating inconsistencies in how requests are accessed, processed, and monitored across departments.

This research will concentrate on the development and implementation of an integrated requisition management system tailored to optimize enterprise workflows. The study will focus on automating core requisition functions. The motivation for this research is informed by practical exposure during an internship experience, where the reliance on multiple disconnected

requisition systems was observed across procurement, maintenance, personnel and IT asset requests. These systems were structured differently, accessed through separate platforms, and often lacked standardization, highlighting a clear gap in the consolidation and management of enterprise-wide requisition processes.

1.2 Problem Statement

The prevailing requisition and approval processes at Isuzu East Africa Limited are characterized by fragmented and semi-manual workflows that lack integration across organizational units, resulting in excessive delays, limited traceability and poor operational visibility. These processes often rely on paper-based or siloed digital systems, leading to duplicate data entry and inconsistent record-keeping. A user making a requisition fills a manual form and experiences complex and prolonged approval chains that extend approval times. The absence of centralized dashboards obscures tracking of a department's requisition activities. Inadequate audit trails undermine compliance, accountability and process optimization. Collectively, these deficiencies reflect a single underlying problem: the absence of an integrated requisition management system capable of standardizing workflows, consolidating data, and supporting transparent approval processes across enterprise functions. There is a critical need for a centralized requisition management solution that streamlines process automation, enhances visibility, and ensures robust policy enforcement in procurement and approval operations (Singh & Chatterjee, 2022; Khatri et al., 2020).

1.3 Aim of Research

The primary aim of this research is to design and develop an integrated, web-based requisition management system to improve workflow efficiency.

1.4 Objectives of the Research

1. To review related requisition management prototypes and systems at both global and local levels to identify relevant limitations.
2. To identify suitable research methodologies and analyze user requirements through techniques such as interviews and document analysis to inform system design.

3. To analyze the existing requisition system and workflow management practices to identify inefficiencies.
4. To design a proposed system architecture that supports multiple requisition categories and structured approval hierarchies.
5. To implement a web-based integrated requisition management system that automates requisition submission, approval routing, status tracking, and LPO generation.
6. To centralize user and catalogue management through administrative controls and provide dashboards for decision-making.
7. To conduct system testing to ensure functionality and reliability.

1.5 Justification of the Research

This research is justified by the persistent operational inefficiencies associated with fragmented and semi-manual requisition and approval processes in many enterprises, particularly within Isuzu East Africa Limited. Observations made during my industrial attachment revealed that requisition activities were often handled through multiple disconnected systems, emails, and paper-based forms, resulting in delayed approvals, duplicated data entry, inconsistent records, and limited visibility across departments. These challenges are consistent with industry findings that show outdated requisition workflows create compliance gaps and approval bottlenecks ultimately increasing operational costs and procurement cycle times (Spendflo, 2025).

Studies on procurement digitalization indicate that automation of requisition workflows can reduce manual effort by up to 40 percent while significantly minimizing errors and approval delays, thereby improving process efficiency and data accuracy (Spendflo, 2025; Singh & Gameti, 2021). Industry-focused research further demonstrates that centralized requisition systems improve approval turnaround times and reduce administrative overhead by automating workflow routing and consolidating requisition data into a single platform (GEP, 2024; Lapala, 2025). Research on system integration emphasizes that organizations increasingly operate complex application landscapes, often exceeding hundreds of systems, which necessitates integration to eliminate silos and improve data flow, security, and scalability (SnapLogic, 2025). These findings directly justify the research objective of designing and implementing an integrated requisition management system to streamline enterprise workflows.

Althabatah et al. (2023) and Santos et al.(2025) emphasize that enterprises achieve greater long-term value when automation and intelligent capabilities are built upon stable, standardized core systems rather than fragmented legacy processes. This supports the decision to focus the present research on developing a robust integrated requisition platform, while deferring analytics-driven and AI-based enhancements to future phases.

Additionally, existing procurement system literature is largely centered on large multinational organizations and developed economies, with limited emphasis on medium-to-large enterprises operating within the East African context. Studies conducted within Kenya indicate that while e-requisition and ERP tools have been adopted, many procurement functions remain partially manual, limiting their overall effectiveness (Ngugi, 2023; Bartai & Kimutai, 2018). By addressing this contextual gap, the research enhances its practical relevance and contributes localized insights into enterprise workflow optimization.

1.6 Scope

This research is specifically limited to Isuzu East Africa Limited and will be conducted over a six-month period. Within this constrained timeline, the study will focus exclusively on automating core requisition workflows related to procurement, merchandise, personnel, and IT asset requisitions. The automated functionalities/features will include:

- Requisition creation: A unified digital form for users to submit requests across all requisition types.
- Multi-level approval workflows: Automated forwarding of requisitions to appropriate approvers through approval hierarchies based on organizational roles and requisition types.
- Requisition status tracking: Tracking of requisition progress, allowing users and approvers to view current statuses such as pending, approved or rejected within the system.
- User and role management: Role-based access control to regulate system access and permissions for requesters, approvers, and administrators.
- Dashboard analytics, reporting and monitoring: Generation of basic reports, including requisition summaries and LPOs.
- Audit trail management: Logging of all requisition activities, including submissions, approvals, rejections, and timestamps, to support accountability and compliance purposes

Excluded from the scope are supplier management modules, contract negotiations, payment processing systems, external system integrations and advanced analytics.

1.7 Report Organization

Chapter 2 reviews academic and industry literature on requisition systems and related workflows, assessing global and local case studies and emerging trends. Chapter 3 details the research methodology encompassing requirement specifications, system design, development, testing, and deployment planning.

Chapter 4 presents the system analysis, including an examination of the existing requisition process, system requirements, feasibility analysis, and the use of structured analysis tools to model the current system. Chapter 5 discusses the design of the proposed system, outlining its architecture, process design, user interactions, data flows, and input-output specifications through various design models and diagrams.

Chapter 6 describes the implementation and testing of the developed system, presenting key system interfaces, testing procedures, evaluation results, and verification of the system requirements. Chapter 7 concludes the report by summarizing the project findings, discussing challenges encountered, presenting recommendations for future improvements, and providing the overall conclusions of the study.

CHAPTER 2: REVIEW OF RELATED WORK

2.1 Chapter Introduction

This chapter reviews scholarly literature and commercial solutions relevant to requisition management systems for enterprise workflow optimization. The review evaluates existing prototypes and commercial products, and discusses emerging design trends.

2.2 History of the Research Topic

Requisition management has been a fundamental component of organizational procurement and resource allocation processes. In early organizational settings, requisition activities were predominantly manual, relying on paper-based forms, physical approvals, and informal communication channels. Procurement inefficiencies often start with small missed approvals and outdated request workflows but they quickly add up to lost time and higher costs. Many growing teams found themselves buried in manual approvals and chasing paper trails just to process basic purchase requests (Spendflo, 2025). While these approaches were sufficient for small-scale operations, they quickly became inefficient as organizations expanded in size and complexity.

With the advancement of information technology in the late twentieth century, enterprises began adopting basic digital tools such as spreadsheets and standalone procurement applications to support requisition processes. According to Deloitte, companies that automate their procurement workflows can cut manual effort by up to 40% and reduce errors significantly (Spendflo, 2025). However, these early digital solutions were often implemented in isolation within departments, resulting in siloed systems that lacked interoperability. Large organizations today operate hundreds of applications across departments, creating fragmented digital environments that undermine workflow efficiency and data consistency (SnapLogic, 2025).

The introduction of ERP systems marked a significant milestone in the evolution of requisition management. ERP platforms enabled centralized databases and integrated business functions, allowing procurement, finance, and operations units to share information more effectively. Studies conducted within manufacturing firms in Nairobi demonstrate that ERP-driven system integration improves procurement efficiency by automating workflows and enabling

cross-departmental data access, thereby enhancing operational performance (Bharatkumar, 2024). Despite these benefits, ERP adoption remains uneven, particularly among medium-sized enterprises, due to high implementation costs and system complexity.

More recently, web-based requisition systems have emerged as flexible alternatives to traditional ERP solutions. Research on web-based material requisition systems in construction supply chains shows that lightweight, web-based platforms can effectively automate requisition submission, approval routing, and tracking using standard web technologies such as HTML, PHP, and MySQL (Adedeji et al., 2021). These systems emphasize accessibility, modularity, and ease of deployment making them suitable for organizations seeking process standardization without full ERP implementation.

Current developments in the field are increasingly oriented towards intelligent and data-driven procurement solutions in the transformation toward industry 4.0 (Spreitzenbarth et al., 2024). Emerging research explores the application of artificial intelligence techniques, such as machine learning and natural language processing, to enhance requisition analysis and bundling. However, scholars emphasize that such advanced capabilities depend on the availability of stable, integrated requisition platforms that can generate reliable and structured data (Bartai & Kimutai, 2018). Consequently, the historical progression of requisition management underscores a gradual shift from manual processes to integrated digital systems, with foundational automation serving as a prerequisite for future intelligent workflow optimization.

2.3 Review of Related Prototypes and Systems

2.3.1 Enterprise Resource Planning (ERP) Systems

Large-scale ERP systems such as **SAP** and **Oracle** have long provided integrated procurement and requisition functionalities within large enterprise environments. Studies on ERP adoption in Kenya indicate that ERP systems enhance procurement efficiency by enabling system integration, data sharing, and workflow automation across organizational units (Bharatkumar, 2024; Ngugi, 2023).

Strengths:

- High level of integration across enterprise functions
- Strong auditability and compliance support
- Centralized data management

Weaknesses:

- High implementation and maintenance costs
- Complexity unsuitable for medium-sized organizations
- Long deployment timelines

These limitations make ERP systems impractical for organizations seeking lightweight, focused requisition solutions.

2.3.2 Commercial Requisition and Procurement Platforms

Modern commercial requisition systems, such as Spendflo (2025) and Lapala (2025), provide centralized digital platforms for managing purchase requisitions and approvals. These systems automate approval routing, improve spend visibility, and reduce administrative workload. Industry reports indicate that such platforms can reduce manual procurement effort by up to 40 percent (GEP, 2024).

Strengths:

- Fast approval cycles
- Improved budgetary control
- User-friendly interfaces

Weaknesses:

- Proprietary systems with limited customization
- Dependence on external vendors
- Limited alignment with localized organizational workflows

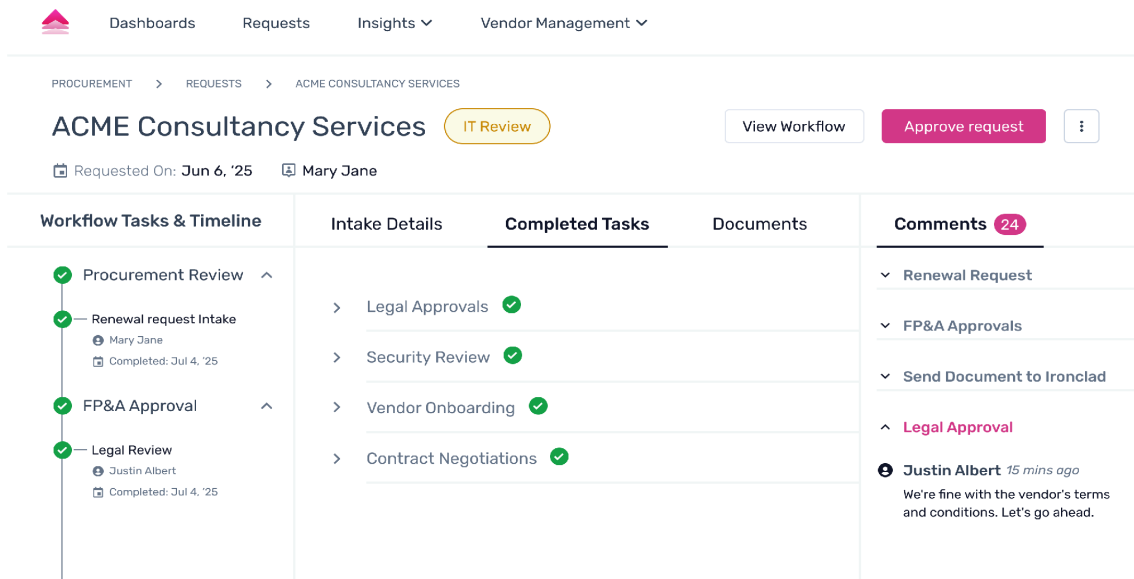


Figure 2.1: Spendflo system screenshot

2.3.3 Public Sector E-Requisition Systems in Kenya

Local studies on e-requisition adoption in Kenya reveal partial digitization of procurement processes within the public sector, specifically North Rift County Assemblies. Bartai and Kimutai (2018) found that e-requisition systems improved order tracking and approval efficiency, however, many procurement functions remained manual.

Strengths:

- Improved procurement performance
- Reduced approval delays

Weaknesses:

- Fragmented implementation
- Incomplete workflow automation

2.3.4 Academic Prototype: Web-Based Material Requisition System

Academic Research has explored web-based requisition systems as alternatives to full-scale platforms. Adedeji et al. (2021) developed a web-based material requisition system for construction businesses using standard web technologies.

Strengths:

- Lightweight and cost-effective
- Built using core web technologies
- Suitable for incremental system development

Weaknesses:

- Limited scalability beyond defined scope
- Minimal integration with external enterprise systems.

REQUISITION Home Log Out

On Site - Dashboard

Welcome, mojisola

[Make Request](#) [Settings](#)

Make Request

Proposed 3-Bedroom Bungalow for Prof. Awosika No. 5 Church Street, Baruwa inside. Ipaja, Lagos

S/N	Item Description	Quantity	Purpose	Delivery Date	
1	Dangote Cement	100	Foundation works	12-Aug-2018	Add More
2	Steel Reinforcement	2000	Foundation works	12-Aug-2018	X

[Request](#)

Figure 2.2: Material Requisition System prototype screenshot

2.3.5 Intelligent and AI-Driven Requisition Systems

Recent studies explore the application of artificial intelligence in procurement. Spreitzenbarth et al. (2024) designed an AI-based purchasing requisition bundling generator that uses machine learning techniques to support procurement decision-making.

Strengths:

- Advanced decision support

Weaknesses:

- High data and infrastructure requirements

- Dependence on stable integrated requisition platforms

This reinforces the need for foundational systems before intelligent enhancements.

2.4 Emerging Trends and Patterns

Requisition and procurement systems are increasingly influenced by digital transformation, system integration, and intelligent automation trends. One prominent trend is the shift from fragmented, department-specific requisition tools toward centralized digital platforms that standardize workflows and consolidate enterprise data (Spendflo, 2025; SnapLogic, 2024).

Another significant trend is the growing emphasis on workflow automation. Industry research highlights that automated approval routing and standardized requisition forms reduce manual intervention, shorten cycle times, and enhance compliance (GEP, 2024; Document Logistix, 2024). Emerging research also highlights the gradual incorporation of analytics and artificial intelligence into procurement systems. AI-drive approaches such as requisition bundling, spend analysis, and decision support are increasingly explored within industry 4.0 contexts (Althabatah et al., 2023; Spreitzenbarth et al., 2024).

Within the East African context, studies indicate a gradual transition towards e-procurement, although adoption remains uneven and partially manual in many organizations (Bartai & Kimutai, 2018; Ngugi, 2023). This pattern underscores the continued relevance of research focused on practical, scalable requisition systems.

2.5 Chapter Summary

This chapter highlighted requisition management's evolution from manual to intelligent systems, evaluated global and regional solutions, and identified emerging trends shaping future developments. Key gaps remain in balancing integration with ease of use and customization for varied enterprise contexts, motivating this research focus.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Chapter Introduction

This chapter shall describe the research methodology that will be adopted to design and develop an integrated, web-based requisition management system for Isuzu East Africa Limited. The chapter will outline the approaches that will be used for data collection, system analysis, design, implementation and testing, drawing directly from operational insights gained during the researcher's industrial attachment at the organization.

3.2 Methodology of Requirement Specification and Data Collection

Requirement specification will employ a qualitative requirements gathering approach, combining interviews, document review and observation. These methods are appropriate for understanding real operational workflows and identifying inefficiencies in existing requisition practices.

3.2.1 Interviews

This is a qualitative data collection method used to obtain detailed insights directly from individuals involved in a process. Interviews will be conducted with the main procurement officer, a departmental head, one HR officer and one finance staff to gain in-depth understanding of workflow challenges, approval hierarchies, and user expectations that may not be captured through quantitative methods. This method has been chosen because it allows flexibility and obtains contextual explanations of problems with the current system. Interview responses will be analyzed to identify common pain points, which will then inform system analysis and design.

3.2.2 Observation

Observation is a qualitative method used to study processes as they occur in their natural setting. Direct observation of current requisition processes at Isuzu will provide a practical understanding of how requisitions are initiated, approved, processed and recorded. Actual practices will be verified and recorded, especially where processes differ from documented procedures. This method has been chosen because it provides first-hand information. Observed workflows will be mapped as a basis for process improvement and system redesign.

3.2.3 Document Review/Analysis

Document review will involve examining existing records and formal documentation to understand procedures and requirements. Current requisition forms, approval templates, and policy documents encountered during the attachment shall be reviewed. Document review has been chosen because it provides authoritative and verifiable information that complements data obtained from interviews and observation. The findings from the document analysis will be used to identify data elements, define system requirements and support system design and implementation.

3.3 Methodology for System Analysis of the Current System

System analysis shall focus on understanding the existing requisition workflows currently in use at Isuzu East Africa Limited and identify areas of inefficiency. A **context diagram** will be developed as a high-level representation of the current system, showing the system boundary, external entities and major data flows. This will be used to define the scope of the existing system and its interaction with users and other organizational units. A **Level 1 Data Flow Diagram (DFD)** will be constructed to decompose the system into major processes, data stores, and data flows, allowing for analysis of data movement and inefficiencies.

3.4 Methodology for System Design (Proposed System)

The system design methodology will focus on translating the analyzed requirements into a structured system model. A **context diagram** will be developed as an illustration of how the new system will interact with users, departments, and external systems while clearly defining system boundaries. A **Level 1 DFD** will be used to represent the major processes of the proposed system and will provide a clear overview of how data flows through the system. A **Level 2 DFD** will further decompose complex processes to ensure all system functions are logically structured and well-defined. Data flow modeling has been chosen because it supports clarity in system logic.

Flowcharts will be developed to describe the logical sequence of operations by modeling approval paths, thereby improving understanding of system behavior. **User interface design** will be approached using standard UI/UX tools to create wireframes and a prototype for validation of system requirements. The wireframes will illustrate system screens such as login, requisition

submission, approval dashboards and reporting interfaces. Prototype screenshots shall be included to demonstrate system navigation and interaction.

Database design will be carried out using an **Entity Relationship Diagram (ERD)**. Entities, attributes, and relationships will be defined based on normalized data from the analysis phase. The ERD will be used to ensure data integrity and ensure future system scalability.

3.5 Methodology for System Implementation

This project will adopt the **Incremental Development Model** as its software development life cycle approach. This model develops the system in small, manageable increments where each increment adds functionality to the previous ones. The modules are developed, tested, and integrated sequentially, allowing for early delivery of core functionality.

3.5.1 Tools and Technology

Front-end:

- HTML5 - to structure and define the content and layout of the user interfaces.
- CSS3 - to style and enhance the visual presentation and responsiveness of the interface.
- JavaScript - to implement client-side interactivity, form validation and dynamic behavior.

Back-end:

- PHP - to handle server-side processing, business logic, authentication and workflow automation,
- MySQL (relational database) - to store, manage, and retrieve requisition data, approval logs, user records and audit trails.

Development Environment:

- Visual Studio Code - primary development environment for writing and debugging code.
- Git & GitHub - for version control and tracking changes.

3.6 Methodology for System Testing

3.6.1 Testing Plan

The process will begin with unit testing, which will be carried out by the developer to confirm that individual modules function correctly. The outputs produced will be compared against expected results to verify correctness. Then integration testing will be conducted to verify that the front-end user interface, PHP backend logic and MySQL database work together as intended and any data flow errors, mismatches or workflow interruptions corrected.

Afterwards system testing will be conducted using a group of ten selected users representing different system roles such as requesters, approvers and system administrators. Each user will be provided with predefined tasks and test scenarios including creating and submitting requisitions, approving or rejecting requests and tracking. User actions and system responses will be observed and recorded to confirm that all system functions operate correctly from start to finish.

3.6.2 Testing Techniques

Testing techniques to be applied include black-box testing, white-box testing, regression testing and security testing. Black-box testing will be applied during user-based system testing and white-box testing will be conducted by the developer, in order to not only verify functionality but also ensure logical correctness and completeness. Regression testing will be carried out after bug fixes or system enhancements to ensure there are no unintended side effects of changes. Security testing will be performed to identify any vulnerabilities such as SQL injection.

3.7 Methodology for System Deployment

3.7.1 Deployment Strategy

A **phased deployment approach** will be adopted, where the system will be rolled out incrementally to different user groups or departments. This strategy has been selected because it will allow for controlled implementation, immediate feedback incorporation, and risk mitigation. Phase 1 will include deployment to one department which will begin using the new system then

phase 2 will extend to additional departments and finally phase 3 achieving organization-wide usage.

3.7.2 Data Migration Plan

Core organizational data required for system operation will be migrated and initialized within the new system. This will include user account details, departmental information, predefined roles and permissions, and approval structures derived from existing policies and records. Data will be verified for accuracy and consistency before import to ensure system reliability. The limited scope of data migration has been chosen to reduce complexity, minimize error, and support a smooth transition from the current system to the proposed solution.

3.8 Chapter Summary

This chapter has outlined the research methodology that will guide the development of the requisition management system for Isuzu East Africa Limited. The methodology for requirements specification through interviews, observation and document analysis to gather qualitative data from stakeholders. The current system will be analyzed using context diagrams, and data flow diagrams. The design phase will employ process modeling, user interface prototyping, and entity-relationship diagrams followed by implementation using an incremental development model. A multi-level testing approach will ensure system quality and deployment will follow a phased strategy. This research methodology framework will ensure the delivery of a functional system that addresses Isuzu East Africa's operational requirements within the six-month project timeline.

CHAPTER 4 : SYSTEM ANALYSIS

4.1 Chapter Introduction

This chapter defines the current fragmented requisition system at Isuzu East Africa Limited, analyzes the strengths and weaknesses of the current system and determines the operational, technical and economic feasibility of the current system. Through analysis of existing processes, this chapter identifies inefficiencies and informs the requirements of the proposed system. Data flow diagrams and flowcharts illustrate how requisitions move through various approval hierarchies. By establishing a baseline of current system performance and limitations, this chapter provides the foundation for designing a solution that genuinely addresses organizational needs.

4.2 Description of the Current System

Isuzu East Africa limited currently manages personnel, procurement, IT assets and merchandise requisitions through a semi-manual process. Each requisition type follows a similar multi-level approval hierarchy but utilizes different form templates stored in filing cabinets, resulting in a fragmented system. Approval notifications and status inquiries are sent via Microsoft Outlook and individual departments maintain separate tracking logs on Excel spreadsheets. This fragmented architecture prevents visibility and creates data silos where information exists in multiple locations with no single source of truth.

The completed form initiates a sequential approval workflow through four mandatory levels:

1. **Department Head** - First-level approval verifying departmental need and budget alignment.
2. **Procurement Head/HR Director** - HR Director only in the case of a Personnel requisition.
3. **Finance Director** - Budget verification and financial viability assessment
4. **Managing Director** - Final strategic approval authority

Each approver physically receives the form, reviews the request, signs or rejects with comments, and forwards to the next approver or returns to the requester. This physical routing process lacks real-time visibility, with requesters having no way to track progress beyond calls or email inquiries.

4.3 Strengths of the Current System

Despite significant limitations, the current requisition system demonstrates some strengths that should be preserved:

1. Current requisition **processes are established and well-understood** by employees across the organization. Despite being manual, employees know whom to approach and what information to provide.
2. **Flexibility in handling requests:** Manual processes allow staff to adjust workflows informally when urgent requests arise.
3. Since much of the process is manual, **system downtime or technical failures have minimal impact.**
4. Physical documentation provides **tangible records** that some stakeholders find reassuring from an **audit and compliance perspective**. Signed physical documents create a clear paper trail that, while inefficient, is familiar and trusted.

4.4 Weaknesses of the Current System

The current requisition system prevents several operational inefficiencies:

1. **Lack of centralized information:** Requisition records are scattered across emails, documents, and spreadsheets, making data retrieval difficult.
2. **Limited Decision Support:** Management lacks access to consolidated analytics and reports for strategic decision-making. Reports require manually gathering data from multiple sources.
3. Manual data entry and document handling **increase the likelihood of errors, duplication, or loss of information.**
4. **Lack of visibility:** Requesters cannot track the approval status, there are no automated notifications and leaders cannot see organization-wide requisition metrics.
5. **Lack of Procurement Integration:** The existing system does not provide an automated mechanism for generating Local Purchase Orders after approvals, resulting in additional manual work.
6. Physical routing and manual reminders lead to **slow approval cycles** as forms must be manually delivered between offices, creating bottlenecks when approvers are unavailable.
7. Risk of **document loss and damage:** Forms can be lost, damaged (spills/tears) and there is no version control meaning changes require new forms or messy annotations. Physical archives degrade over time.

8. Despite the organization having Microsoft 365 infrastructure there is **technology underutilization** and requisition systems that don't integrate with ERP and procurement systems.

4.5 Feasibility Study and Conclusion

A feasibility study was conducted to determine whether implementing a new requisition management system would be practical and beneficial for the organization.

4.5.1 Operational Feasibility

Operational feasibility evaluates whether the proposed system will improve organizational processes and be accepted by users.

The proposed system will streamline requisition creation, approval, and tracking processes through a centralized platform. Employees will be able to submit requisitions digitally, while managers can approve requests electronically. Given the organization's existing use of digital tools such as email and spreadsheets, staff are expected to adapt easily to the proposed system.

Operational Feasibility Conclusion

The proposed system is operationally feasible because it directly addresses the shortcomings of the current process, supports existing organizational workflows, improves transparency and accountability, and provides users with tools that simplify requisition management and procurement activities.

4.5.2 Technical Feasibility

Technical feasibility assesses whether the organization has the technological capability to support the proposed system.

The proposed system will be developed as a **web-based application**, using widely supported technologies such as **PHP, MySQL, HTML, CSS, and JavaScript**. These technologies are

compatible with the organization's existing IT infrastructure and require minimal additional hardware.

The system can be hosted on the organization's internal server or deployed on a secure web server, ensuring accessibility across departments.

Technical Feasibility Conclusion

The proposed system is technically feasible because the required hardware, software, network infrastructure, and technical expertise are readily available. The selected technologies are reliable, cost-effective, scalable, and capable of supporting the functional requirements of the system.

4.5.3 Economic Feasibility

Economic feasibility evaluates whether the benefits gained from implementing the proposed system outweigh the associated development and operational costs.

The current manual requisition process incurs recurring costs associated with paper-based documentation, printing, physical document handling, administrative effort, delayed approvals, and inefficient procurement tracking. These inefficiencies consume employee time and reduce organizational productivity.

The proposed Rezon system automates requisition processing, approval routing, notifications, reporting, audit logging, and Local Purchase Order generation, resulting in significant operational savings and productivity gains.

TABLE 4.1: ESTIMATED ANNUAL COST COMPARISON

Cost Item	Current Manual Process (KES/Year)	Proposed System (KES/Year)
Printing and stationery	120,000	15,000
Paper storage and filing	60,000	5,000
Administrative processing time	350,000	100,000
Follow-up communication costs	80,000	10,000
Delays and process inefficiencies	250,000	50,000
System maintenance	0	50,000
Server and hosting costs	0	30,000
Total Annual Cost	860,000	260,000

The estimated annual savings achieved through implementation of the proposed system are:

Annual Savings = 860,000 – 260,000 = KES 600,000

This indicates that the organization can recover the development cost within the first year of operation while continuing to realize savings in subsequent years.

Economic Feasibility Conclusion

The proposed system is economically feasible because the expected operational savings significantly exceed the implementation and maintenance costs. The estimated return on investment demonstrates that the system provides substantial financial benefits while improving organizational efficiency and accountability.

4.6 System Analysis

System analysis involves examining the structure and operations of the existing requisition management process in order to understand how information flows within the organization. This analysis helps to identify system components, users, data flows and interactions between different actors.

4.6.1 Structured System Analysis and Design (SSAD)

Structured System Analysis and Design is a methodological approach used to analyze information systems by breaking them down into smaller components in order to understand how data flows within the system. A **context diagram** and **data flow diagram (DFD)** are used to illustrate the boundaries of the system and flow of information.

4.6.2 Context Diagram

A high-level representation of the semi-manual personnel requisition process at Isuzu East Africa. The primary external entities include *the requesting employee, department head, directors of the relevant departments and finally the Managing Director*. The context diagram highlights the major data flows, including submission of paper form, requests for review and approval decisions.

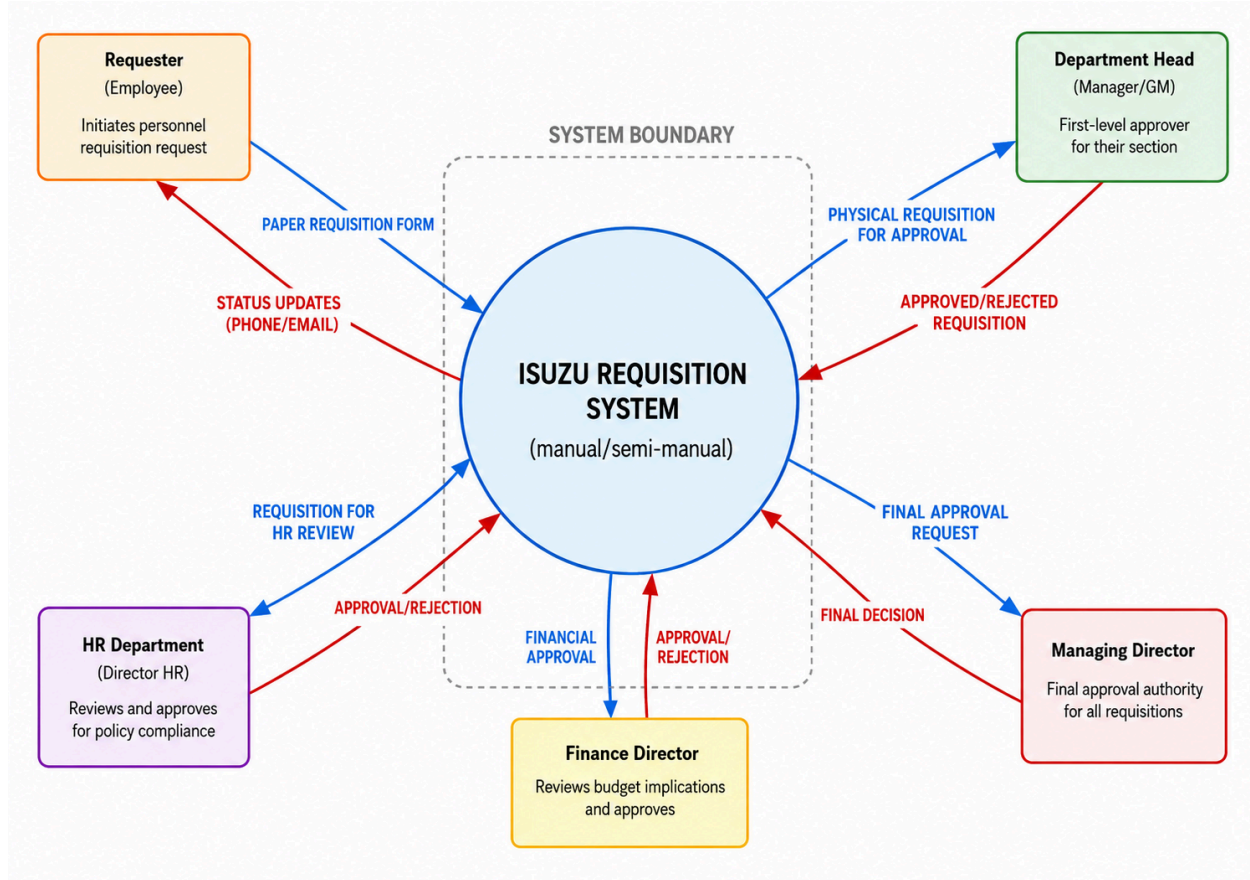


Figure 4.1: Current System Context Diagram

4.6.3 Level 1 Data Flow Diagram (DFD)

The Data Flow Diagram provides a more detailed representation of how data moves through different processes within the system. First is requisition *submission* from the requester, *approval routing/processing* and *generation of manual reports* that can be shared with Management.

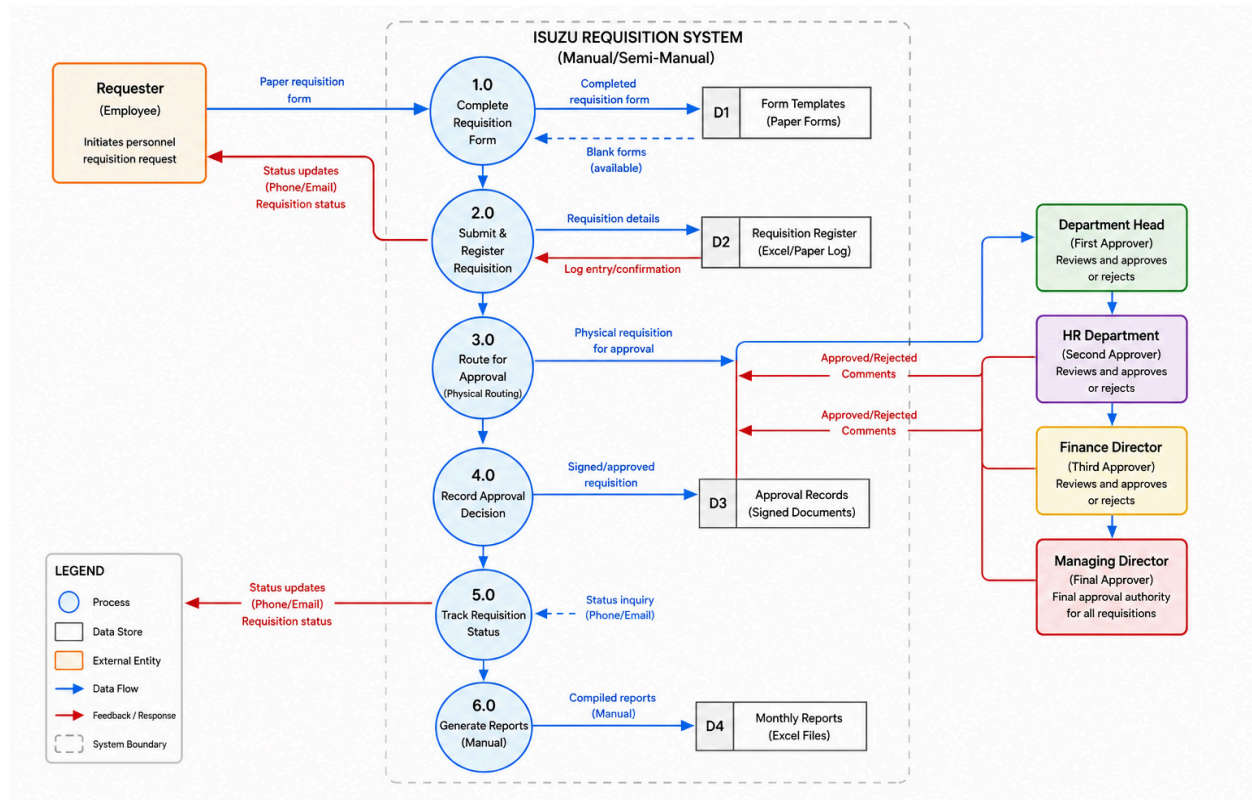


Figure 4.2: Current System Level DFD

4.6.4 System Users and Actors

The requisition management system involves several users who provide inputs to the system and receive outputs based on their responsibilities.

TABLE 4.2: USER ROLES, INPUTS, AND OUTPUTS

User/Actor	Role/Responsibility	Inputs to the System	Outputs from the System
Requester (Employee)	Initiates requisition requests when resources or personnel are needed.	Requisition details, item description, quantity, department information, justification.	Submission confirmation, requisition status updates, approval/rejection notifications.

Department Head	Reviews and validates departmental resource requirements as the first-level approver.	Approval/rejection decision, comments, approval date.	Updated requisition status, approval notifications, requisition details for review.
HR Director <i>(where applicable)</i>	Reviews personnel requisitions to ensure alignment with workforce planning requirements.	Approval/rejection decision, comments, approval date.	Position details, departmental comments, approval notifications.
Procurement Head <i>(where applicable)</i>	Processes approved requisitions and initiates procurement procedures.	Approval/rejection decision, comments, approval date.	Procurement updates, purchase processing information, notifications.
Finance Director	Evaluates budget availability and financial feasibility of requisitions.	Approval/rejection decision, budget-related comments, approval date.	Cost estimates, budget information, approval notifications.
Managing Director	Provides final approval for requisitions.	Approval/rejection decision, comments, approval date.	Fully reviewed requisition, complete approval history, notifications.
IT Systems Administrator	Maintains the system, manages configurations, monitors logs, and supports users.	Log updates, reminder alerts, configuration updates.	Requisition logs, audit records, system reports, monitoring information.

4.6.5 Process Flowchart

Process flowcharts provide detailed step-by-step visualization of key workflows in the current system.

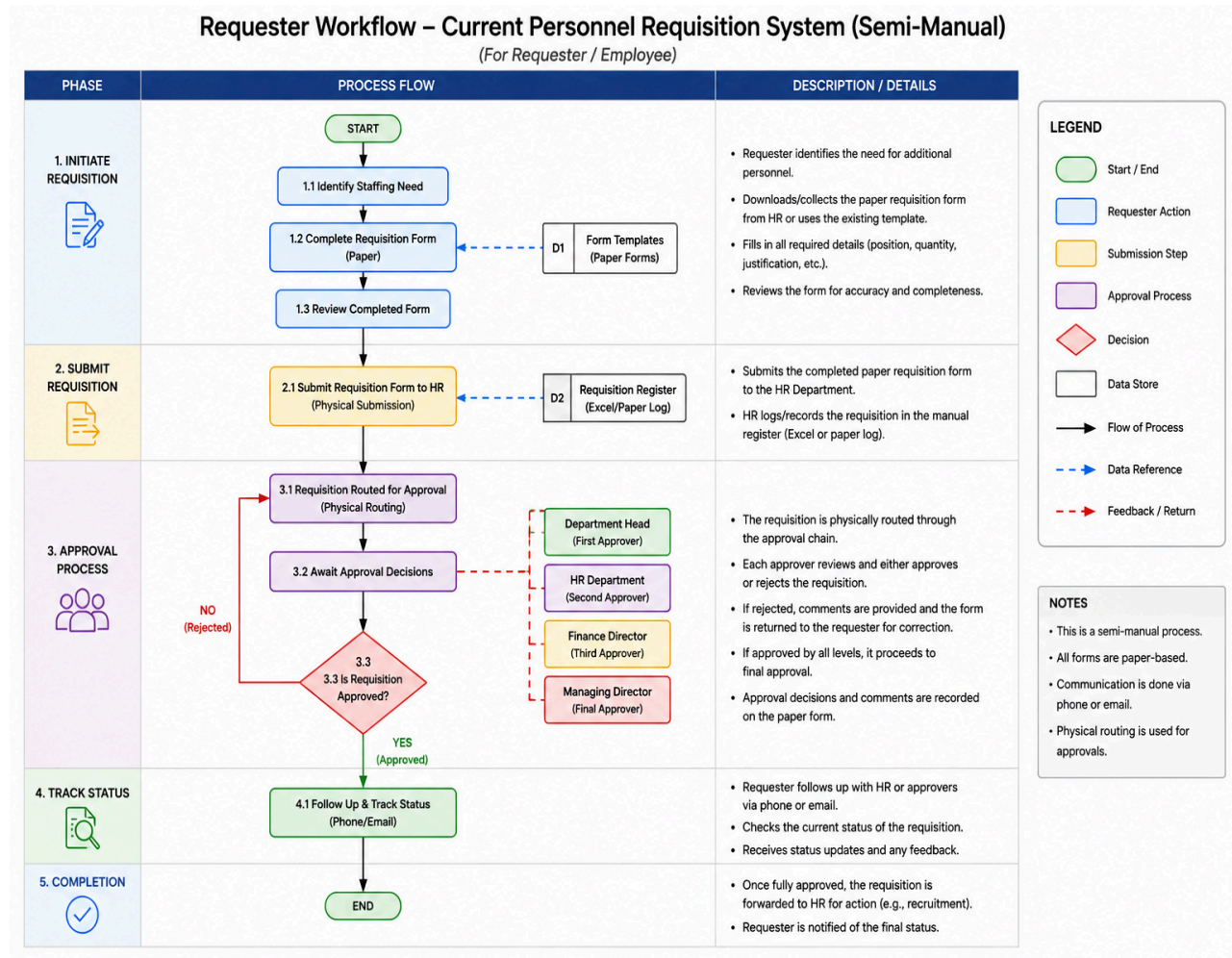


Figure 4.3: Current Process Flowchart - Requester

4.7 Chapter Summary

This chapter has comprehensively analyzed the current requisition management system at Isuzu East Africa, revealing a semi-manual and fragmented process for handling different requisitions. The strengths of the current system such as clear approval hierarchies and established processes provide a solid foundation for automation. However, weaknesses like manual inefficiencies, limited visibility and reporting gaps demonstrate the need for an integrated solution. Feasibility analysis confirms that the proposed system is feasible across all dimensions and requirements analysis has translated identified problems into specific functional, non-functional and data requirements or specification that will guide the proposed system's design. The next chapter will detail the system design that addresses these requirements.

CHAPTER 5: SYSTEM DESIGN

5.1 Chapter Introduction

This chapter presents the design of the proposed requisition management system developed to address the limitations identified in the current system. The chapter outlines the structure, components, and operational processes of the new system, focusing on how it improves efficiency, transparency, and accountability in requisition handling.

The design phase translates the requirements identified in Chapter 4 into a structured system architecture and process models. It provides a blueprint for system implementation by defining system components, user interactions, and data flow within the system.

Additionally, this chapter presents diagrams such as system architecture, flowcharts, activity diagrams, and sequence diagrams to visually represent the functionality and interactions within the proposed system.

5.2 Description of the Proposed System

The proposed system is a **web-based requisition management system** designed to automate and centralize the process of requisition submission, approval, procurement, and tracking at Isuzu East Africa Limited.

The system allows employees to submit requisition requests electronically through a user-friendly interface. Once submitted, the system automatically routes the request to the appropriate department manager for review and approval. Approved requisitions are then forwarded to the procurement department for processing.

The system provides real-time tracking capabilities, enabling users to monitor the status of their requisitions without the need for manual follow-ups. Notifications are automatically generated to inform users of approvals, rejections, and procurement updates.

All requisition data is stored in a centralized database, ensuring easy retrieval, improved data integrity, and enhanced reporting capabilities. The system also incorporates role-based access control to ensure that users can only access functionalities relevant to their roles.

5.3 Strengths of the Proposed System

The proposed system offers several advantages over the current system:

1. Centralized data management

All requisition-related data is stored in a single database, improving accessibility and reducing data redundancy.

2. Improved efficiency

Automation of workflows reduces delays associated with manual approvals and document handling.

3. Real-time tracking and transparency

Users can monitor the status of their requisitions at any stage, improving visibility and accountability.

4. Enhanced accuracy

Automated data entry and validation reduce errors associated with manual processes.

5. Better reporting and decision-making

The system enables generation of reports that support management in analyzing procurement trends and performance.

6. Improved communication

Automated notifications eliminate the need for constant follow-ups via email or phone.

5.4 Weaknesses of the Proposed System

Despite its advantages, the proposed system has some limitations:

1. Initial implementation cost

The organization may incur costs related to system development, deployment, and training.

2. Dependence on technology

System availability depends on network connectivity and server reliability.

3. User resistance to change

Some employees may be reluctant to adopt the new system, especially if they are accustomed to manual processes.

4. Maintenance requirements

The system will require regular updates, monitoring, and technical support to ensure optimal performance.

5.5 Requirements Analysis

Requirements analysis identifies the system functionalities and performance characteristics necessary to meet organizational needs.

5.5.1 Functional Requirements

Functional requirements describe the specific services and operations that the proposed Integrated Requisition and Approval System (Reqon) must provide to its users. These requirements were derived from the weaknesses identified in the existing requisition process and the operational needs of Isuzu East Africa Limited.

1. Requisition Creation and Submission

The system shall provide digital requisition forms that enable users to create and submit requisition requests electronically. The system shall automatically generate unique requisition numbers and validate all mandatory fields before submission.

2. Approval Routing and Workflow Management

The system shall automatically route requisitions through predefined approval hierarchies based on requisition type, department, and approval level. Approvers shall be able to review, approve, reject, or return requisitions while maintaining workflow integrity.

3. Status Tracking and Approval Visibility

The system shall allow users to track the status of submitted requisitions in real time. Users shall be able to view approval history, current approval stage, comments, and final decisions associated with each requisition.

4. Local Purchase Order (LPO) Generation

The system shall generate Local Purchase Orders (LPOs) for requisitions that have successfully completed all required approval stages. Generated LPOs shall be assigned unique LPO numbers and stored for future reference.

5. Search and Filtering

The system shall provide search and filtering functionality to enable users to quickly locate requisitions and approval records based on specified criteria.

5.5.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance expectations, security controls, and operational characteristics of the proposed system.

1. Performance

The system shall provide acceptable response times under normal operating conditions to ensure an efficient user experience. Dashboard and navigation pages shall load within three seconds, while requisition submissions shall be processed within five seconds. Search and filter operations shall return results within three seconds to enable users to retrieve information quickly. The system shall also support concurrent access by multiple users without significant degradation in performance and shall generate Local Purchase Orders (LPOs) and reports within ten seconds under normal operating conditions.

2. Usability

The system shall provide a user-friendly interface that minimizes training requirements and promotes user adoption across the organization. The interface shall utilize intuitive navigation menus and consistent page layouts to ensure ease of use. Clear labels, instructions, and validation messages shall be provided to guide users during system interaction. Common tasks such as requisition creation and approval shall require minimal steps to complete, while digital requisition forms shall closely resemble the organization's existing physical forms to reduce the learning curve. Additionally, the system shall provide immediate feedback after user actions such as submissions, approvals, and rejections.

3. Reliability

The system shall operate consistently and accurately while maintaining the integrity of organizational data. It shall prevent duplicate requisition submissions by generating unique requisition identifiers and validating transactions before processing. The system shall maintain transaction consistency throughout workflow processing and preserve all submitted data in the event of unexpected failures. Accurate approval histories and audit logs shall be maintained for accountability purposes, and the system shall recover gracefully from processing errors without corrupting stored information.

4. Security

The system shall protect organizational data from unauthorized access, modification, and disclosure. User authentication shall be enforced before access to any system functionality is granted, and Role-Based Access Control (RBAC) shall be implemented to restrict access according to assigned user roles and responsibilities. User passwords shall be encrypted using secure hashing algorithms before storage in the database. Furthermore, all user inputs shall be validated and sanitized prior to processing, and parameterized database queries shall be used to protect against SQL Injection attacks. The system shall maintain audit logs of critical user activities and automatically terminate inactive sessions after a predefined period of inactivity to minimize unauthorized access risks.

5. Scalability

The system shall be designed to accommodate future organizational growth in terms of users, requisition volumes, and functional requirements. The dynamic three-step digital requisition form shall support the addition of new requisition categories without requiring significant system redesign. The system architecture shall also allow future integration with external systems such as email services, inventory management systems, supplier management platforms, and Enterprise Resource Planning (ERP) systems, ensuring that the solution remains adaptable to evolving business needs.

5.6 Conceptual Architecture of the Proposed System

The proposed system adopts a **three-tier architecture**, consisting of the presentation layer, application layer, and database layer.

I. Presentation Layer

This is the user interface layer through which users interact with the system. It is accessible via web browsers and provides interfaces for submitting requisitions, approving requests, and tracking status.

II. Application Layer

This layer contains the system's business logic and handles processing of user requests. It manages workflows such as requisition validation, approval routing, and procurement processing.

III. Database Layer

This layer is responsible for storing all system data, including user information, requisition records, approval history, and procurement details. A relational database management system ensures data integrity and consistency.

The architecture ensures separation of concerns, making the system more scalable, secure, and easier to maintain.

5.7 Process Design

Process design defines how the system operates by illustrating workflows and interactions between system components and users. It ensures that all system activities are logically organized and aligned with the system requirements.

5.7.1 Proposed System Context Diagram

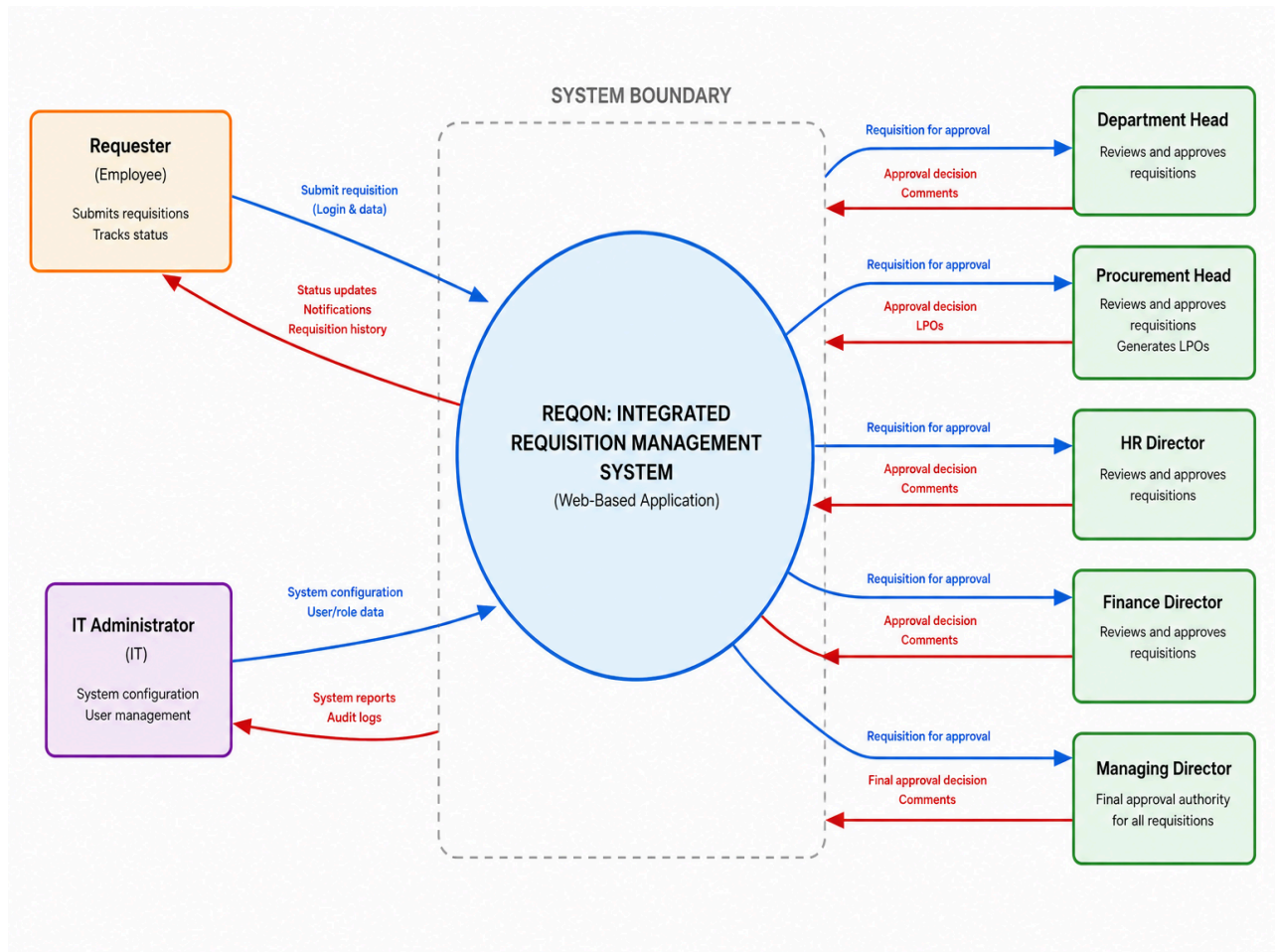


Figure 5.1: Proposed System Context Diagram

5.7.2 Level 1 DFD

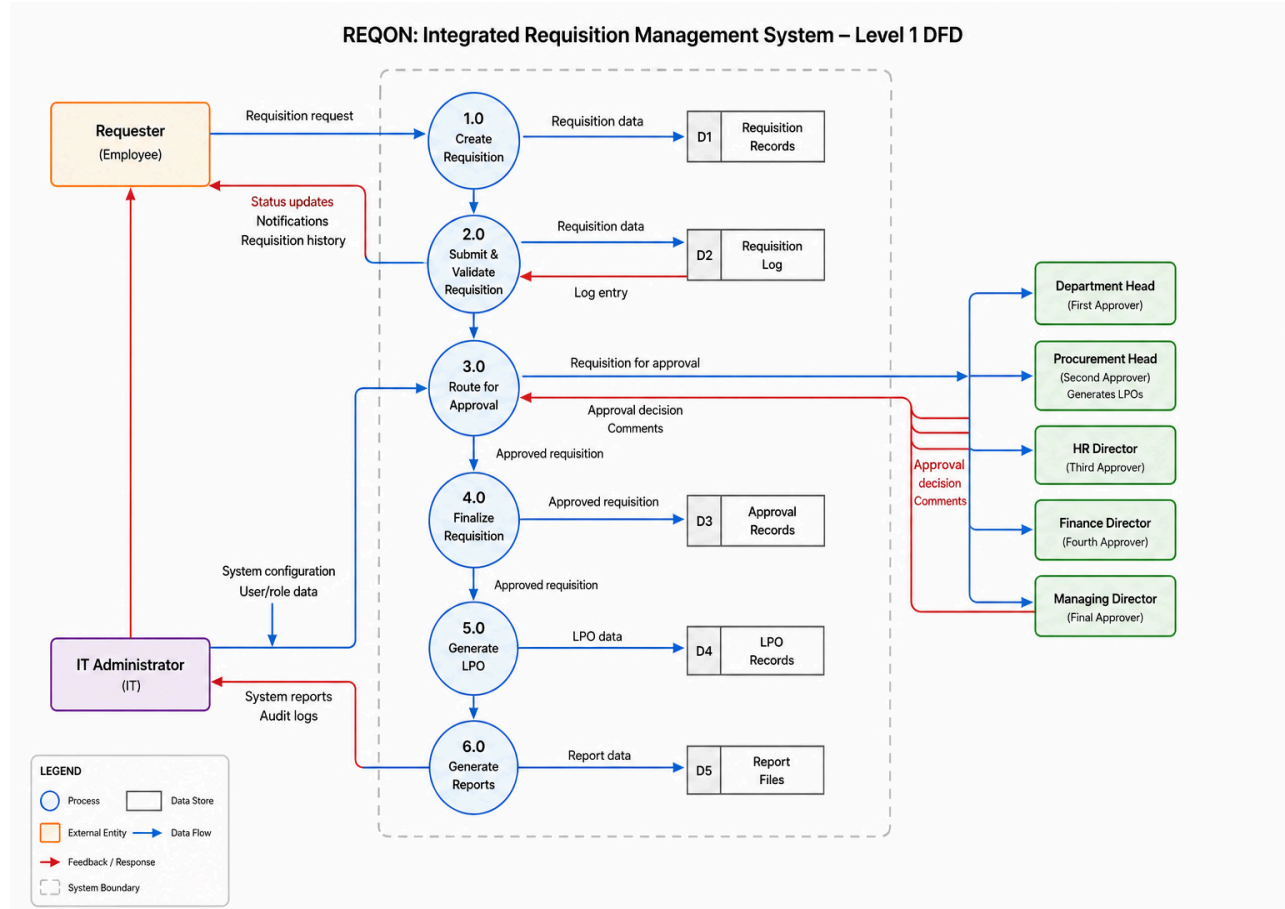


Figure 5.2: Proposed System Level 1 DFD

5.7.3 Flowcharts

Flowcharts are used to represent the step-by-step processes within the proposed system. They illustrate how requisitions are created, approved, processed, and completed.

In the proposed system, the process begins when a user logs into the system and submits a requisition request. The system validates the input and automatically forwards the request to the appropriate manager for approval.

If the request is approved, it is sent to the procurement department for processing. If rejected, the system notifies the user with feedback. Once procurement is completed, the system updates the requisition status and notifies the user.

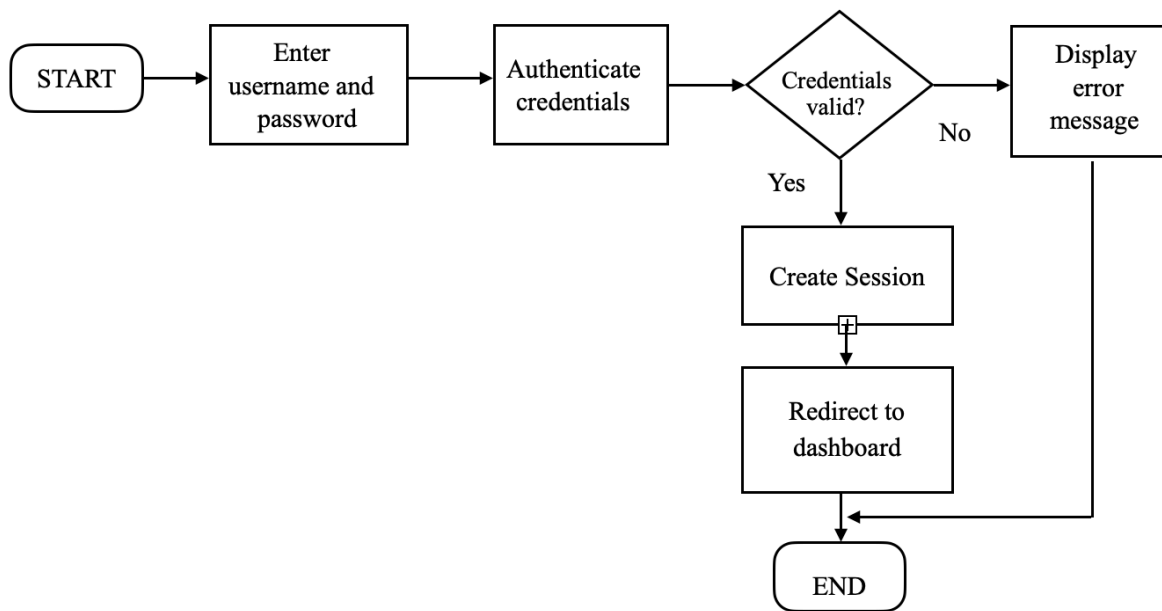


Figure 5.3: User Login Process Flowchart

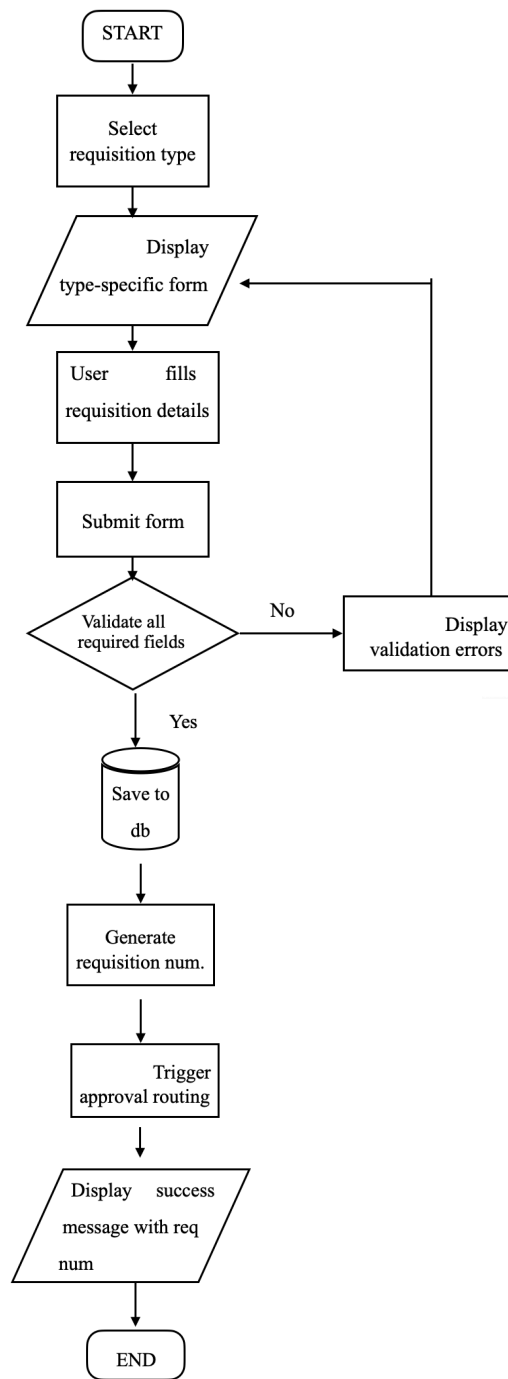


Figure 5.4: Requisition Submission Process

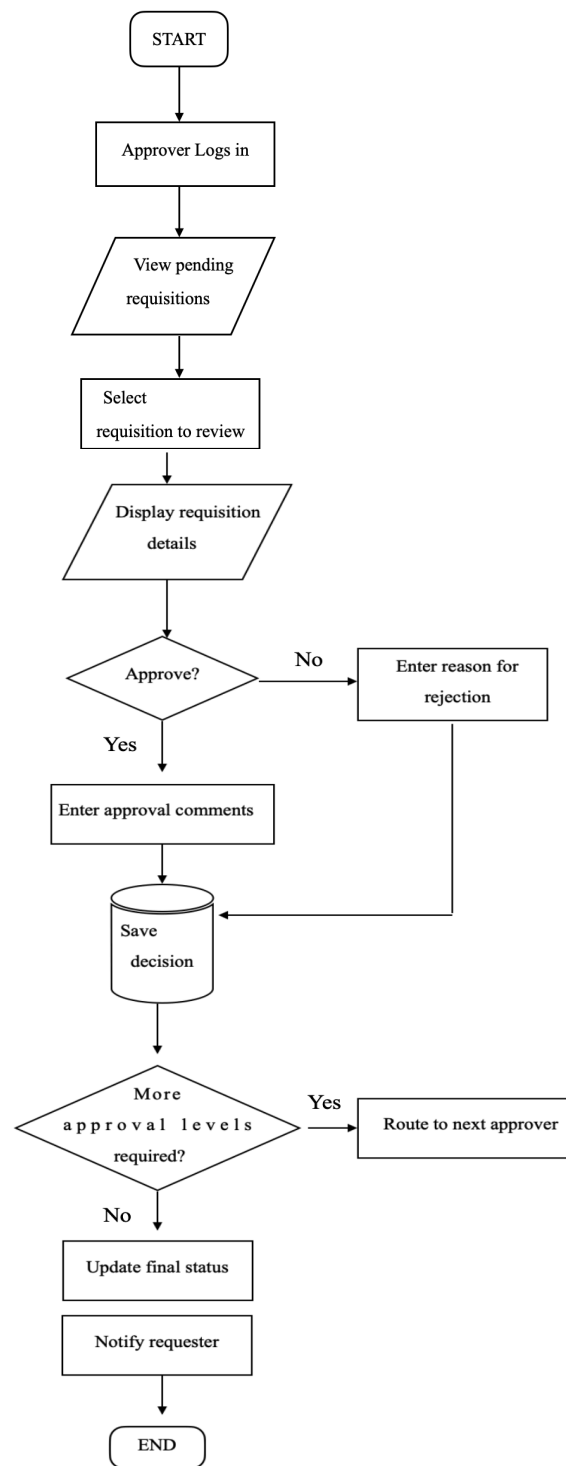


Figure 5.5: Approval Decision Process

5.7.4 Sequence Diagram

The sequence diagram illustrates the interaction between system components and users over time. It shows how messages are exchanged between actors and the system to complete a requisition process.

The sequence begins with the employee submitting a requisition request through the user interface. The system processes the request and sends it to the manager for approval. The manager responds with an approval or rejection decision.

If approved, the system forwards the request to the procurement officer, who processes the requisition and generates an LPO updating the system.

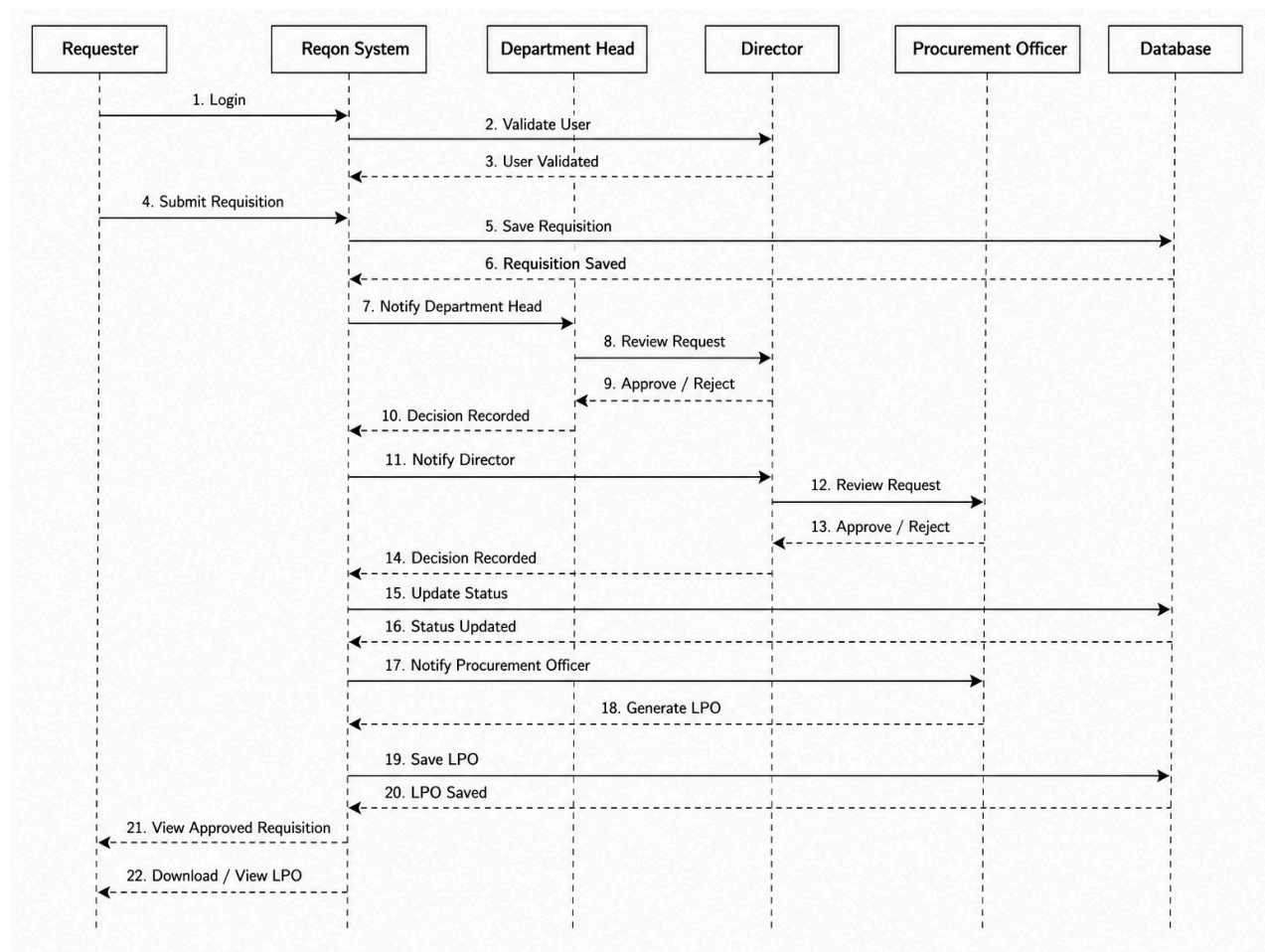


Figure 5.6: Sequence Diagram

5.8 Entity - Relationship Diagram (ERD)

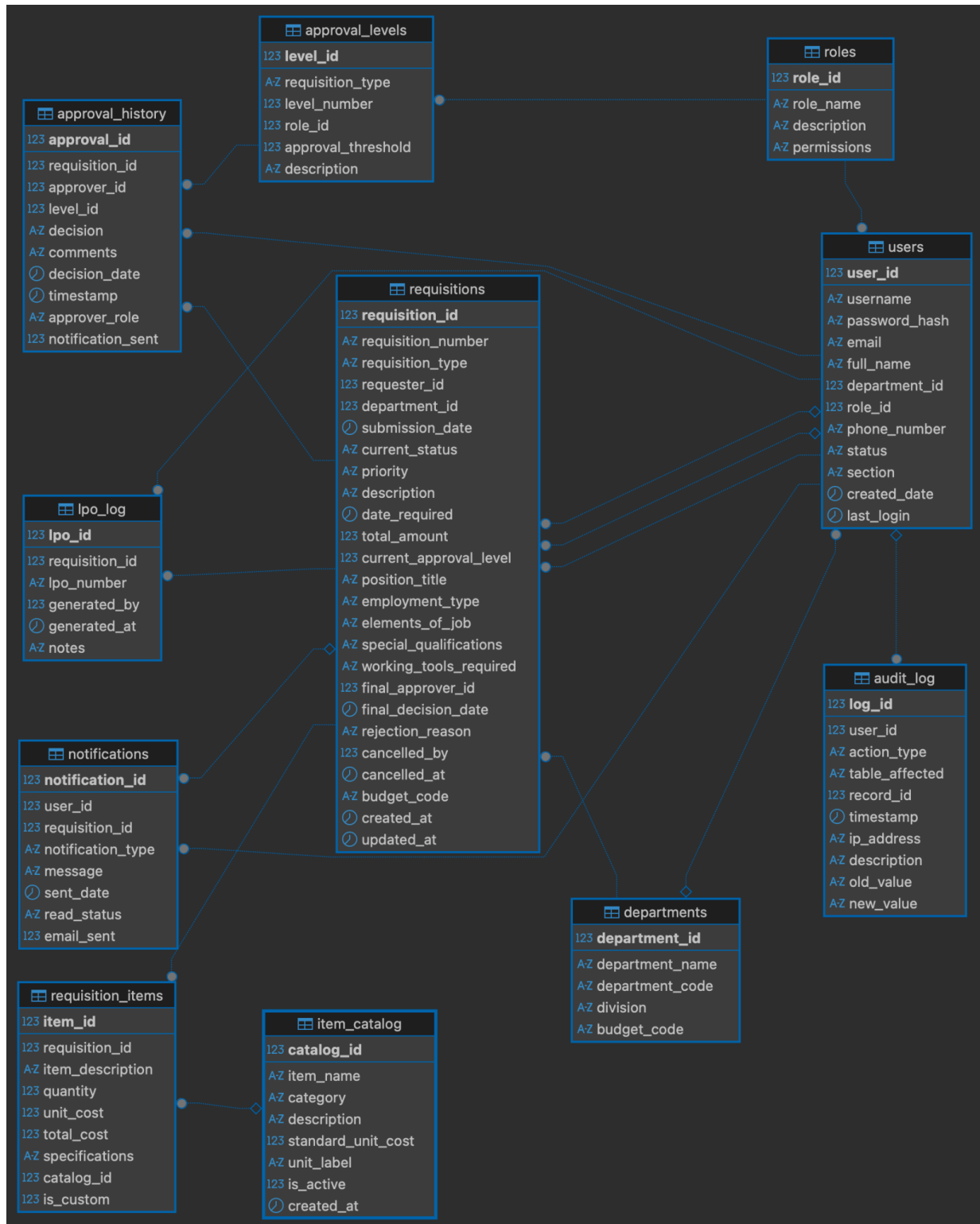


Figure 5.7: Entity Relationship Diagram (ERD)

5.9 Data Dictionary

The data dictionary provides detailed descriptions of the database entities, their attributes, relationships, and purposes within the Integrated Requisition and Approval System (Regon). It serves as a reference for system development, maintenance, and future enhancements by ensuring consistency in data usage throughout the application.

5.9.1 Users Table

Stores user account information and authentication details. Each user belongs to a department and is assigned a role that determines system access permissions.

TABLE 5.1: USERS TABLE

Column Name	PK/FK	Data Type	Nullable	Description
user_id	PK	INTEGER	No	Unique identifier for each user.
username		VARCHAR(80)	No	User login username.
password_hash		VARCHAR(255)	No	Encrypted user password.
email		VARCHAR(120)	No	User email address.
full_name		VARCHAR(120)	No	Full name of the user.
department_id	FK	INTEGER	Yes	References DEPARTMENTS.department_id.
role_id	FK	INTEGER	No	References ROLES.role_id.
phone_number		VARCHAR(20)	Yes	User contact number.
status		ENUM	No	User account status (active/inactive).
section		VARCHAR(100)	Yes	User's organizational section.
created_date		DATETIME	No	Date account was created.

last_login		DATETIME	Yes	User's most recent login timestamp.
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5.9.2 Departments Table

Stores departmental information used for requisition routing and reporting.

TABLE 5.2: DEPARTMENTS TABLE

Column Name	PK/FK	Data Type	Nullable	Description
department_id	PK	INTEGER	No	Unique department identifier.
department_name		VARCHAR(100)	No	Department name.
department_code		VARCHAR(20)	No	Department code.
division		VARCHAR(100)	Yes	Organizational division.
budget_code		VARCHAR(30)	Yes	Department budget code.

5.9.3 Roles Table

Defines user roles and associated permissions within the system.

TABLE 5.3: ROLES TABLE

Column Name	PK/FK	Data Type	Nullable	Description
role_id	PK	INTEGER	No	Unique role identifier.
role_name		VARCHAR(50)	No	Name of the role.
description		VARCHAR(255)	Yes	Role description.
permissions		JSON	Yes	Stored role permissions.

5.9.4 Requisitions Table

Stores all requisition requests submitted through the system.

TABLE 5.4: REQUISITIONS TABLE

Column Name	PK/FK	Data Type	Nullable	Description
requisition_id	PK	INTEGER	No	Unique requisition identifier.
requisition_number		VARCHAR(20)	No	System-generated requisition number.
requisition_type		ENUM	No	Type of requisition submitted.
requester_id	FK	INTEGER	No	References USERS.user_id.
department_id	FK	INTEGER	No	References DEPARTMENTS.department_id.
submission_date		DATETIME	No	Date requisition was submitted.
current_status		ENUM	No	Current requisition status.
priority		ENUM	No	Request priority level.
description		TEXT	Yes	Requisition justification.
date_required		DATE	Yes	Required completion date.
total_amount		DECIMAL(15,2)	No	Total requisition cost.
current_approval_level		TINYINT	No	Current workflow stage.
final_approver_id	FK	INTEGER	Yes	Final approving officer.
final_decision_date		DATETIME	Yes	Date of final decision.

5.9.5 Requisition_Items Table

Stores individual items associated with a requisition.

TABLE 5.5: REQUISITION_ITEMS TABLE

Column Name	PK/FK	Data Type	Nullable	Description
item_id	PK	INTEGER	No	Unique item identifier.
requisition_id	FK	INTEGER	No	References REQUISITIONS.requisition_id.
item_description		VARCHAR(255)	No	Description of requested item.
quantity		INTEGER	No	Requested quantity.
unit_cost		DECIMAL(15, 2)	No	Cost per unit.
total_cost		DECIMAL(15, 2)	No	Computed total cost.
specifications		TEXT	Yes	Additional item specifications.
catalog_id	FK	INTEGER	Yes	References ITEM_CATALOG.catalog_id.
is_custom		BOOLEAN	No	Indicates custom item entry.

5.9.6 Item_Catalog Table

Stores standard items available for requisition.

TABLE 5.6: ITEM_CATALOG TABLE

Column Name	PK/FK	Data Type	Nullable	Description
catalog_id	PK	INTEGER	No	Unique catalogue item identifier.
item_name		VARCHAR(255)	No	Item name.
category		ENUM	No	Item category.
description		TEXT	Yes	Item description.

standard_unit_cost		DECIMAL(15,2)	No	Standard item cost.
unit_label		VARCHAR(30)	Yes	Unit of measurement.
is_active		BOOLEAN	No	Indicates active catalogue status.
created_at		DATETIME	No	Date item was created.

5.9.7 Approval Levels

Stores workflow approval hierarchy definitions used to determine the sequence of approvers for different requisition categories and organizational structures.

TABLE 5.7: APPROVAL_LEVELS TABLE

Column Name	PK/FK	Data Type	Nullable	Description
level_id	PK	INTEGER	No	Unique approval level identifier.
requisition_type		VARCHAR(50)	No	Requisition category associated with the approval level.
level_number		INTEGER	No	Sequence number indicating the approval order.
role_id	FK	INTEGER	No	References ROLES.role_id. Specifies the role responsible for approval at this level.

level_name		VARCHAR(100)	No	Descriptive name of the approval stage.
is_active		BOOLEAN	No	Indicates whether the approval level is active.
created_at		DATETIME	No	Date and time the approval level was created.

5.9.8 Approval_History Table

Stores all approval decisions made during the requisition approval process. Provides a complete audit trail of approvals, rejections, and workflow progression.

TABLE 5.8: APPROVAL_HISTORY TABLE

Column Name	PK/FK	Data Type	Nullable	Description
history_id	PK	INTEGER	No	Unique approval history record identifier.
requisition_id	FK	INTEGER	No	References REQUISITIONS.requisition_id.
approver_id	FK	INTEGER	No	References USERS.user_id.

level_id	FK	INTEGER	No	References APPROVAL_LEVELS.level_id.
action_taken		ENUM	No	Approval action performed (Approved, Rejected, Returned).
comments		TEXT	Yes	Approver remarks or rejection reason.
action_date		DATETIME	No	Date and time the action was performed.
approval_status		VARCHAR(20)	No	Status after the action was recorded.

5.9.9 Notifications Table

Stores system-generated notifications sent to users regarding requisition activities, approvals, rejections, LPO generation, and other system events.

TABLE 5.9: NOTIFICATIONS TABLE

Column Name	PK/FK	Data Type	Nullable	Description
notification_id	PK	INTEGER	No	Unique notification identifier.
user_id	FK	INTEGER	No	References USERS.user_id. Recipient of the notification.

requisition_id	FK	INTEGER	Yes	References REQUISITIONS.requisition_id. Associated requisition if applicable.
title		VARCHAR(150)	No	Notification title or subject.
message		TEXT	No	Detailed notification content.
notification_type		VARCHAR(50)	No	Type of notification (Approval, Rejection, LPO, System Alert, etc.).
is_read		BOOLEAN	No	Indicates whether the notification has been viewed.
created_at		DATETIME	No	Date and time the notification was generated.

5.9.10 Lpo_log table

Stores Local Purchase Orders generated from fully approved requisitions. Tracks procurement processing and LPO issuance history.

TABLE 5.10: LPO_LOG TABLE

Column Name	PK/FK	Data Type	Nullable	Description
lpo_id	PK	INTEGER	No	Unique LPO identifier.

requisition_id	FK	INTEGER	No	References REQUISITIONS.requisition_id.
lpo_number		VARCHAR(50)	No	System-generated Local Purchase Order number.
generated_by	FK	INTEGER	No	References USERS.user_id. Procurement officer who generated the LPO.
supplier_name		VARCHAR(255))	Yes	Supplier associated with the LPO.
total_amount		DECIMAL(15, 2)	No	Total LPO amount including applicable taxes.
vat_amount		DECIMAL(15, 2)	Yes	Calculated VAT amount.
generation_date		DATETIME	No	Date and time the LPO was generated.
status		VARCHAR(20)	No	Current LPO status (Generated, Printed, Issued, Completed).

5.9.11 Audit_Log Table

Stores records of significant user actions performed within the system. Supports accountability, monitoring, security reviews, and compliance requirements.

TABLE 5.11: AUDIT_LOG TABLE

Column Name	PK/FK	Data Type	Nullable	Description
audit_id	PK	INTEGER	No	Unique audit log identifier.
user_id	FK	INTEGER	No	References USERS.user_id. User who performed the action.
action_type		VARCHAR(100)	No	Type of action performed.
entity_type		VARCHAR(100)	No	System entity affected (Requisition, User, LPO, Catalogue Item, etc.).
entity_id		INTEGER	Yes	Identifier of the affected record.
description		TEXT	No	Detailed description of the action performed.
ip_address		VARCHAR(45)	Yes	IP address from which the action originated.
timestamp		DATETIME	No	Date and time the action occurred.

5.10 Proposed System Prototype

SCREEN 1: LOGIN PAGE

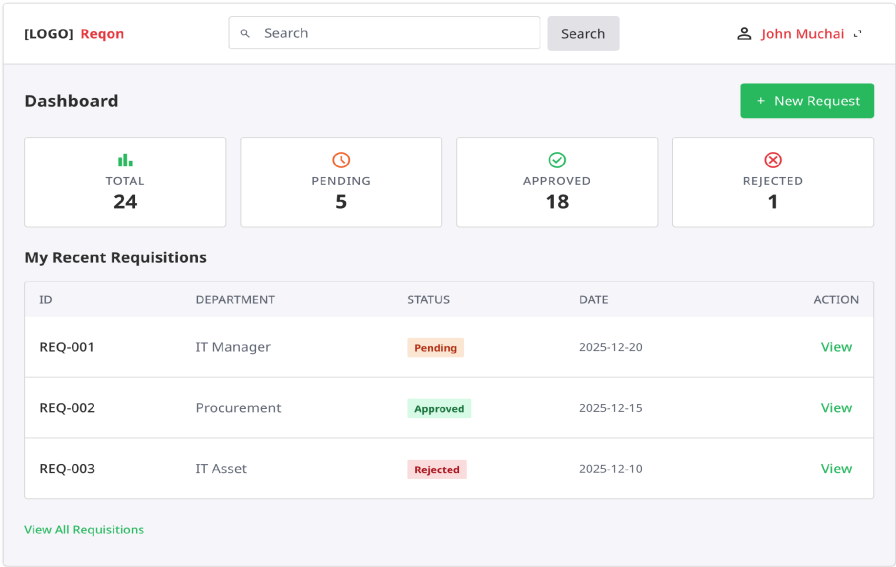
ISUZU EA

Repon
Requisition Management System

Email



Screen 1: Login Page Design



Screen 2: Dashboard Page Design

[LOGO] Regon

Dashboard > New Requisition

Create New Requisition

Step 1 of 3: Requisition Type

(progress indicator)

Select Requisition Type *

Personnel

Position Title *

Department *

Select Department

Section *

Select Section

Screen 3: Digital Requisition Form

← Back

REQ-005 Details

⋮ Actions

🕒 Status: Pending

📁 Type: IT Asset

! Priority: High

📅 Date: 2025-12-27

REQUISITION DETAILS

Position:IT Manager

Department:IT

Section:Support

Required:Jan 15

Items:5 laptops

Total:KES 500,000

Justification:
New Team Expansion

APPROVAL PROGRESS

●●○●○ Level 2 of 4

🟢 Dept Head (Approved)

→ HR Director (Pending)

○ Finance Director

○ Managing Director

COMMENTS HISTORY

Elizabeth (Dept Head)

Dec 27, 2025 - 10:30 AM

"Approved. Urgent need for new members."

🔒 APPROVERS ONLY

Your Decision

Add Comments (optional)

Type your comments here...

✔ Approve

✖ Reject

Screen 4:
Requisition Details
Page (Report)

5.11 Chapter Summary

This chapter presented the design of the proposed requisition management system, outlining its structure, functionalities, and operational processes. The system was designed to address the inefficiencies identified in the current system by introducing automation, centralized data management, and real-time tracking capabilities.

The chapter described the system architecture and process design using tools such as flowcharts, activity diagrams, and sequence diagrams to illustrate system operations and interactions between users.

Additionally, the inputs and outputs of the system were identified, demonstrating how data is captured, processed, and transformed into useful information that supports decision-making and improves system efficiency.

Overall, the proposed system design provides a comprehensive framework for implementing an effective requisition management system. The next chapter will focus on the **implementation and testing** of the system.

CHAPTER 6: SYSTEM IMPLEMENTATION AND TESTING

6.1 Chapter Introduction

This chapter presents the implementation and testing of the Requisition Management System developed for Isuzu East Africa Limited. The chapter demonstrates how the system was implemented to address the challenges identified during system analysis and design. It also presents screenshots of key system interfaces and discusses the testing and evaluation activities conducted to verify that the system meets the specified requirements.

6.2 System Implementation

The requisition management system was implemented as a web-based application to automate requisition processing, approval workflows, procurement support, and management reporting.

The system was developed using PHP for server-side processing, MySQL for database management, HTML and CSS for user interface design, and JavaScript for client-side interactivity and validation.

It supports multiple user roles including Requesters, Department Heads, Directors, the Managing Director, Procurement Personnel, and System Administrators. Through role-based access control, each user is provided with functionalities relevant to their responsibilities.

The implemented system enables users to submit requisitions electronically, track approval progress, receive notifications, and monitor requisition status in real time. Approved requisitions are automatically routed to an LPO Queue where the Procurement Department Head generates Local Purchase Orders (LPOs). The system also provides dashboard analytics, audit logging, user management, and catalogue management functionalities to support operational efficiency and informed decision-making.

6.3 System Screenshots

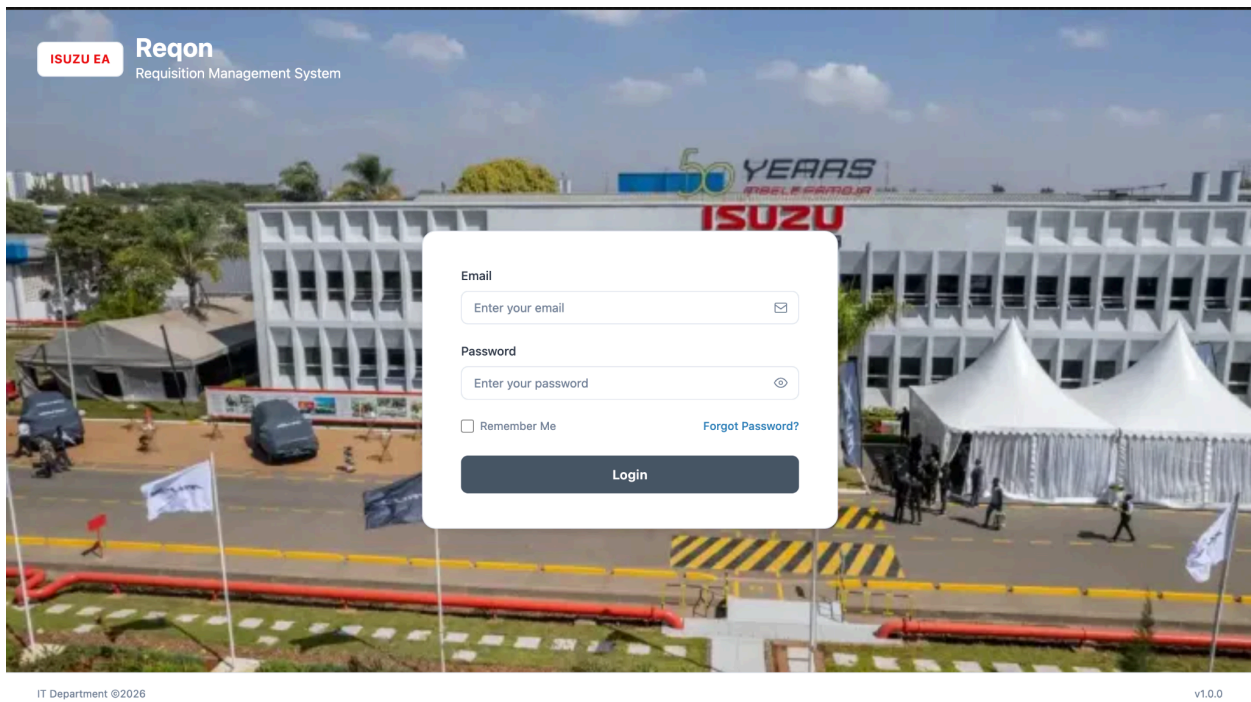


Figure 6.1: Login Interface

The login page provides secure authentication and grants access to users based on their assigned roles within the organization.

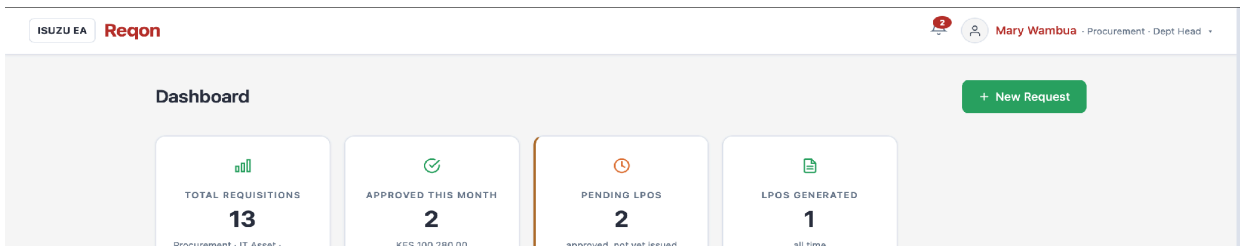


Figure 6.2 Dashboard screen

The dashboard provides users with an overview of requisition statistics, pending approvals, expenditure trends, procurement activities, and key performance indicators relevant to their role.

The screenshot shows the 'Create New Requisition' form in the Reqon system. The form is part of a 3-step process: Step 1 of 3 (Type & Details), Step 2 (Items & Details), and Step 3 (Review & Submit). The first step is active. The form includes the following fields and options:

- Requisition Type ***: A dropdown menu with the option 'Choose type'.
- Department ***: A dropdown menu with the option 'Select Department'.
- Date Required ***: A date input field showing '06/10/2026'.
- Priority ***: Three radio button options: 'Low', 'Medium' (selected), and 'High'.

At the bottom of the form, there are two buttons: 'Cancel' and 'Next Step →'.

Figure 6.3: 3-Step Requisition Form

This interface enables users to create and submit requisitions by selecting catalogue items and providing the necessary request information.

LPO LPO-015 · REQ-015 · Human Resources
Print / Save as PDF

Isuzu East Africa Limited
Requisition Management System — Reqon

LOCAL PURCHASE ORDER
LPO-015

LPO NUMBER
LPO-015
REQUISITION REFERENCE
REQ-015
REQUESTING DEPARTMENT
Human Resources
REQUESTED BY
Sharon Kitavi
APPROVED ON
10/06/2026

ISSUE DATE
10/06/2026
DATE REQUIRED
03/06/2026
BUDGET CODE
BDG-HR-2026
FINAL APPROVER
James Ngugi
REQUISITION TYPE
IT Asset

#	DESCRIPTION	QTY	UNIT COST (KES)	TOTAL (KES)
1	Dell Latitude 5540 Laptop	1	100,000.00	100,000.00
Total Amount				KES 100,000.00

Subtotal
VAT (16%)
Grand Total (incl. VAT)

KES 100,000.00
KES 16,000.00
KES 116,000.00

JUSTIFICATION / DESCRIPTION
Needed for new IT team

TERMS & CONDITIONS
1. This LPO is valid only when signed by an authorised signatory of Isuzu East Africa Limited.

Figure 6.4: LPO Generated by the System after approvals

6.4 Testing Plan

Testing was conducted after the implementation phase to verify that the Integrated Requisition and Approval System (Reqon) functioned according to the specified requirements and supported the complete requisition lifecycle from request creation to Local Purchase Order (LPO) generation. The testing process combined technical verification with end-user evaluation to ensure both functional correctness and usability.

A working prototype of the system was deployed in a controlled testing environment and demonstrated to **eight (8) representative users** drawn from the major stakeholder groups identified during requirements analysis. The participants consisted of **three (3) Requesters, two (2) Department Heads, one (1) Director, one (1) Procurement Officer, and one (1) System Administrator**.

During the demonstration, participants were guided through realistic business scenarios that reflected actual organizational processes. These scenarios included creating requisitions,

submitting requests for approval, reviewing and approving requisitions, rejecting requests, generating Local Purchase Orders (LPOs), managing users, maintaining catalogue items, and reviewing dashboard reports. Following the demonstration, participants were allowed to interact with the system independently and perform tasks corresponding to their respective organizational roles.

A total of **five (5) test scenarios** were executed during the evaluation:

1. User Authentication and Login
2. Requisition Creation and Submission
3. Approval and Rejection Processing
4. LPO Generation
5. Dashboard Reporting and User Administration

All five scenarios were completed successfully, resulting in a **100% test success rate**. The system correctly routed requisitions through the approval hierarchy, generated notifications, maintained approval histories, and successfully produced Local Purchase Orders for approved requests.

Feedback was collected through interviews and direct observation. The evaluation revealed that:

- **87.5% (7 out of 8 users)** found the system easy to learn and navigate.
- **100% (8 out of 8 users)** agreed that requisition tracking and visibility were significantly improved compared to the existing process.
- **87.5% (7 out of 8 users)** indicated that the automated approval workflow reduced the effort required to process requisitions.
- **75% (6 out of 8 users)** considered the dashboard reports useful for monitoring requisition activities and expenditure trends.
- **100% (8 out of 8 users)** agreed that the LPO generation functionality improved procurement processing and reduced administrative effort.

Observations made during testing indicated that users were able to complete assigned tasks with minimal assistance after a brief system demonstration. Minor suggestions were received

regarding interface improvements and additional reporting options, which were documented as potential future enhancements.

6.4.1 Unit Testing

Unit testing was conducted on individual modules to verify that each component performed its intended function independently before integration with other modules.

The following modules were tested individually:

- User Authentication Module
- Requisition Creation and Submission Module
- Approval Workflow Module
- Notification Module
- Dashboard Reporting Module
- User Management Module
- Catalogue Management Module
- Local Purchase Order (LPO) Generation Module
- Audit Logging Module

For each module, valid and invalid input scenarios were executed to verify correct system responses. Authentication testing verified successful login using valid credentials and appropriate error messages for invalid credentials. Requisition submission testing verified mandatory field validation and the successful generation of unique requisition numbers.

6.4.2 Integration Testing

Integration testing was performed to verify that individual modules interacted correctly and that data flowed accurately between the presentation layer, application logic, and database.

The following integrations were specifically tested:

- Requisition submission and approval workflow integration.
- Approval workflow and notification generation integration.
- Requisition approval and LPO generation integration.

- User management and role-based access control integration.

Testing confirmed that requisition data was correctly transmitted between modules and that workflow transitions occurred without data loss, duplication, or inconsistency.

6.4.3 System Testing

System testing was conducted on the fully integrated application to validate complete business processes and ensure that all functional requirements were satisfied.

The following end-to-end workflows were tested:

- User authentication and role-based access control.
- Requisition creation and submission.
- Multi-level approval routing.
- Requisition rejection and resubmission.
- Notification generation and delivery.
- Local Purchase Order generation.
- Dashboard reporting and analytics.
- User and catalogue management.

The objective of system testing was to ensure that the application functioned correctly under normal operating conditions and that all components worked together as a complete requisition and procurement management solution.

TABLE 6.1: SAMPLE TEST CASES

Test Case	Expected Result	Actual Result	Outcome
User Login	Successful authentication and dashboard access	User redirected to role-specific dashboard	Pass
Requisition Submission	Request saved and routed for approval	Requisition submitted and routed successfully	Pass

Approval Processing	Request advances through workflow	Approval recorded and next approver notified	Pass
Requisition Rejection	Rejection reason recorded and requester notified	Requisition rejected and notification generated	Pass
LPO Generation	LPO generated from approved requisition	LPO created successfully with unique LPO number	Pass
User Creation	New user account created successfully	User account added and assigned role	Pass
Catalogue Item Creation	New catalogue item available for requisition	Item added successfully	Pass
Dashboard Reporting	Statistics displayed accurately	Dashboard metrics reflected system data	Pass

This structure surfaces the user-based testing immediately, which is exactly what most supervisors and examiners are looking for when they ask "How did you test with end users?" rather than hiding it under a separate UAT subsection.

6.5 Evaluation Plan

The developed system was evaluated against the objectives established in Chapter One to determine the extent to which the project achieved its intended outcomes. Evaluation was conducted using system testing results, user feedback obtained during User Acceptance Testing, and observations made throughout system usage.

6.5.1 Evaluation of Objective 1: Automate Requisition Creation and Submission

The first objective sought to eliminate reliance on manual requisition forms and email-based submissions by providing a centralized electronic platform for creating and managing requisitions.

Evaluation results showed that users were able to create, submit, and track requisitions entirely within the system. The automatic generation of requisition numbers and centralized storage of requisition records eliminated duplication and improved record management. During testing, all submitted requisitions were successfully captured and routed to the appropriate approval stage without manual intervention.

Evaluation Outcome: Objective successfully achieved.

6.5.2 Evaluation of Objective 2: Automate Approval Routing and Workflow Management

The second objective aimed to streamline approval processes through automated workflow routing and approval management.

Testing confirmed that requisitions were automatically routed to the correct approvers according to predefined approval hierarchies. Approvers were able to review, approve, reject, or return requisitions while maintaining complete workflow visibility. Approval actions were recorded accurately, and notifications were generated at each stage of the process.

Evaluation Outcome: Objective successfully achieved.

6.5.3 Evaluation of Objective 3: Improve Visibility and Tracking of Requisitions

The third objective sought to provide users with real-time visibility into the status of their requisitions and improve process transparency.

Evaluation demonstrated that users could monitor approval progress, view approval history, identify pending approvers, and receive notifications regarding workflow changes. During User Acceptance Testing, all participants agreed that the system significantly improved visibility compared to the previous process.

Evaluation Outcome: Objective successfully achieved.

6.5.4 Evaluation of Objective 4: Improve Procurement Support through LPO Generation

The fourth objective focused on extending requisition processing into procurement activities through automated Local Purchase Order generation.

Testing confirmed that fully approved requisitions were successfully routed to the LPO Queue, where the Procurement Officer could generate and manage Local Purchase Orders directly from the system. Generated LPOs were assigned unique identifiers and stored for future reference.

All participants involved in procurement activities agreed that this functionality reduced administrative effort and improved procurement workflow continuity.

Evaluation Outcome: Objective successfully achieved.

6.5.5 Evaluation of Objective 5: Support Management Decision-Making

The final objective sought to provide management with meaningful information to support operational monitoring and strategic decision-making.

The dashboard analytics module successfully provided information on requisition volumes, approval performance, expenditure trends, pending approvals, procurement activities, and LPO generation statistics. During testing, managers were able to access real-time insights through the dashboard, enabling improved monitoring of organizational activities.

Evaluation Outcome: Objective successfully achieved.

Overall Evaluation

The evaluation results indicate that the Integrated Requisition and Approval System (Reqon) successfully achieved all project objectives. During testing, all five test scenarios were completed successfully, resulting in a **100% functional success rate**. User feedback further demonstrated high levels of satisfaction with the system's usability, workflow automation, visibility features, procurement support capabilities, and reporting functionality.

Compared to the previous manual process, the system reduced dependence on email-based communication, improved requisition traceability, automated approval routing, enabled centralized record management, and introduced Local Purchase Order generation capabilities.

The inclusion of dashboard analytics further enhanced management oversight by providing real-time access to operational and procurement information.

Based on the testing outcomes and user feedback, the developed system was found to be suitable for deployment and capable of supporting efficient requisition, approval, and procurement management within Isuzu East Africa Limited.

6.6 Chapter Summary

This chapter presented the implementation and testing of the Requisition Management System. The implemented solution successfully automated requisition workflows, approval processes, procurement support activities, and Local Purchase Order generation. Testing results confirmed that the system meets the functional and non-functional requirements identified during analysis and design. The evaluation demonstrated improvements in efficiency, transparency, accountability, and decision-making within the requisition management process.

CHAPTER 7: CONCLUSIONS, FINDINGS AND RECOMMENDATIONS

7.1 Chapter Introduction

This chapter presents the findings, conclusions, challenges encountered, and recommendations arising from the design, development, implementation, and evaluation of the Requisition Management System.

7.2 Conclusion

The purpose of this project was to investigate the challenges associated with the requisition and approval processes at Isuzu East Africa Limited and develop an automated solution capable of improving efficiency, transparency, accountability, and decision-making. The study was motivated by the limitations observed in the existing requisition process, which relied heavily on email communication, manual approvals, fragmented record keeping, and limited visibility into the progress of requests. These challenges often resulted in delayed approvals, difficulties in tracking requisition status, duplicated effort, and insufficient management information for procurement planning and expenditure monitoring.

The literature review established that organizations increasingly rely on integrated information systems to automate approval workflows, improve operational efficiency, and strengthen organizational control. Concepts such as workflow automation, role-based access control, audit logging, centralized reporting, and procurement integration were identified as best practices for modern requisition management systems. These findings guided the design and development of the proposed solution.

System analysis further confirmed that the existing requisition process at Isuzu East Africa Limited suffered from operational inefficiencies due to fragmented communication channels, lack of workflow visibility, and limited reporting capabilities. The feasibility study demonstrated that the development and implementation of the proposed system was operationally, technically, and economically viable, with the anticipated benefits outweighing implementation and maintenance costs.

Based on the findings of this study, it can be concluded that the Integrated Requisition Management System (Reqon) successfully addresses the challenges associated with the existing requisition process at Isuzu East Africa Limited. The system provides a centralized platform for requisition submission, approval management, procurement support, reporting, and administrative control, thereby eliminating many of the inefficiencies associated with manual and fragmented workflows.

7.3 Key Findings

1. **The existing requisition process was fragmented and inefficient.**

The analysis established that requisition activities were managed through a combination of emails, manual approvals, and isolated record-keeping methods, resulting in poor coordination, delayed approvals, and limited visibility into workflow progress.

2. **Manual approval processes contributed significantly to operational delays.**

The absence of automated routing mechanisms required users to rely on manual follow-ups and communication, creating bottlenecks and increasing approval turnaround times.

3. **Users lacked real-time visibility into requisition status.**

Requesters were often unable to determine where their requisitions were within the approval chain, leading to uncertainty and frequent inquiries regarding approval progress.

4. **The organization lacked centralized requisition records and reporting capabilities.**

Historical requisition information was dispersed across different communication channels, making retrieval, monitoring, auditing, and reporting difficult and time-consuming.

5. **Workflow automation significantly improved process efficiency.**

The implemented system successfully automated requisition routing through predefined approval hierarchies, reducing manual intervention and ensuring consistency throughout the approval process.

6. **Role-Based Access Control improved system security and accountability.**

By restricting access according to user roles and responsibilities, the system ensured proper segregation of duties and enhanced data security.

7. Integrated notifications improved communication among stakeholders.

The notification module enabled users to receive timely updates on requisition submissions, approvals, rejections, and workflow progression, reducing dependence on manual communication.

8. Audit logging enhanced transparency and traceability.

The system maintained detailed records of approvals, rejections, user actions, and workflow transitions, thereby improving accountability and supporting compliance requirements.

9. Dashboard analytics improved management visibility and decision-making.

The system provided real-time insights into requisition volumes, approval performance, departmental expenditure trends, pending approvals, and procurement activities, enabling management to make more informed decisions.

10. The integration of Local Purchase Order (LPO) generation streamlined procurement operations.

Unlike traditional requisition systems that end at the approval stage, Reqon extends the workflow by supporting the generation of LPOs from fully approved requisitions, creating a seamless transition from approval to procurement execution.

11. Administrative management features improved maintainability and operational control.

The system allows administrators to manage users and catalogue items through dedicated interfaces, reducing reliance on direct database administration and simplifying system maintenance.

12. The developed system successfully met the project objectives.

Testing and evaluation confirmed that the system fulfilled both the functional and non-functional requirements identified during analysis while improving efficiency, transparency, accountability, and organizational control.

7.4 Challenges Encountered

Several challenges were encountered during the development of the project:

1. **Requirements Gathering:** Understanding and documenting approval workflows across different departments required extensive consultation and validation.
2. **Database Design:** Designing a database structure capable of supporting multiple user roles, approval levels, notifications, audit logs, and procurement activities required careful planning.
3. **Workflow Automation:** Implementing dynamic approval routing and ensuring accurate workflow progression across different requisition categories presented significant technical challenges.
4. **LPO Integration:** Extending the system beyond requisition approval to include Local Purchase Order generation required additional workflow design and testing.
5. **Testing and Validation:** Testing multiple user roles and approval scenarios was time-consuming but necessary to ensure system reliability and accuracy.

7.5 Future Recommendations

Although the system successfully achieves its intended objectives, several enhancements can be considered in future versions:

1. **Email Integration:** Implement automated email notifications for approvals, rejections, and LPO generation events.
2. **Mobile Optimization:** Develop a mobile-responsive interface or dedicated mobile application to improve accessibility.
3. **Supplier Management:** Introduce supplier registration, performance tracking, and supplier evaluation functionalities.
4. **Inventory Integration:** Integrate the system with inventory management systems to improve stock visibility and procurement planning.
5. **Advanced Analytics:** Incorporate predictive analytics and expenditure forecasting to support strategic procurement decisions.
6. **Enterprise Integration:** Enable integration with Enterprise Resource Planning (ERP) systems to facilitate organization-wide data sharing and process automation.

7.6 Summary

The Integrated Requisition and Approval System (Recon) successfully transformed a fragmented and largely manual requisition process into a centralized, automated, and transparent platform. By supporting requisition management, approval automation, Local Purchase Order generation, dashboard analytics, and administrative controls, the system has improved operational efficiency, accountability, and decision-making. The project provides a strong foundation for future enhancements and demonstrates the effectiveness of digital solutions in modernizing organizational procurement and approval processes.

APPENDICES

Appendix A: Project Schedule (Gantt chart)

TABLE 1: PROJECT SCHEDULE

Activities	Start & End Dates	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mar 2026	Apr 2026	May 2026	June 2026
Proposal Development	Oct 1 - Nov 30									
Requirements & Analysis	Nov 1 - Dec 20									
System Design	Dec 20 - Jan 1									
System Implementation	Jan 1 - May 28									
System Testing	May 28 - Jun 1									
Documentation	Oct 1 - Jun 15									

Note: Some tasks may overlap or run concurrently to optimize the project timeline.

Appendix B: Project Budget and Resources

TABLE 2: PROJECT BUDGET

Category	Sub-Category	Justification/Details	Estimated Cost (KES)	Source/Vendor	Subscription
<i>Hardware</i>	Development Laptop	To support development	100000.00	Apple Store	One-time
<i>Software</i>	Operating System/Tools	MacOS (Free), VS Code (Free), Git (Free)	0.00	Open Source	One-time
<i>Materials & Operations</i>	Printing/Binding	Final proposal and appendix printing	2500.00	Local Print Shop	One-time
	Transport	Travel for final presentation/submission	500.00	Personal Expense	One-time
	Internet/Communication	Allocated portion of home internet for development/research	7500.00	ISP	6 months
<i>Contingency Fund (10%)</i>	Unforeseen Expenses	Buffer for unexpected software licenses or additional costs	500.00	N/A	One-time
TOTAL PROJECT BUDGET			110,000.00 KES		

Appendix C: Sample Interview Guide

Introduction (5 minutes)

“ Good morning/afternoon. My name is Sharon Kitavi, a final-year Bachelor of Science in Information Technology student conducting research on the Design and Development of an Integrated Requisition Management System (Reqon) for Isuzu East Africa Limited.

The purpose of this interview is to understand the current requisition and approval processes, identify existing challenges, and gather requirements for the proposed system. The information provided will be used strictly for academic purposes and will be treated confidentially. Participation is voluntary, and the interview is expected to take approximately **20–30 minutes**.”

Section A: Current Requisition Process

1. Can you describe how requisitions are currently initiated, approved, and processed from your perspective?
2. What role do you play in the requisition workflow?
3. What tools, systems, or documents do you currently use to manage requisitions?

Section B: Challenges and Pain Points

4. What are the major challenges you experience with the current requisition process?
5. Have you encountered delays or failures in requisition approvals? If yes, what caused them?
6. How do you currently track the status of pending requisitions?
7. Where do you observe the most significant bottlenecks in the approval process?

Section C: System Requirements

8. What key features would you expect from an automated requisition management system?
9. How should approval routing differ across requisition types (e.g., procurement, maintenance, personnel)?
10. What information is essential for you when reviewing or approving a requisition?
11. What types of reports or summaries would support your decision-making?

Section D: Organizational and Technical Considerations

12. What policies or compliance requirements should the system enforce?
13. How should user access, roles, and permissions be managed?

14. How important are audit trails and accountability records in requisition processing?

Section E: Implementation and Adoption

15. What concerns or expectations do you have regarding the implementation of a new requisition system?

Appendix D: Sample Code

/approvals/action.php - Processes an approve/reject decision then directs to next approver.

```
<?php

require_once __DIR__ . '/../config/config.php';
require_once __DIR__ . '/../includes/auth.php';

requireLogin();

// Only accept POST
if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . '/approvals/queue.php');
}

$user = currentUser();
$userLevel = getRoleLevel($user);

// Check the user has an approval role
if ($userLevel === 0) {
    setFlash('error', 'You do not have permission to approve or reject requisitions.');
```

```
    redirect(BASE_URL . '/dashboard.php');
}

// Collect and validate inputs
$reqId = (int)post('requisition_id');
$action = post('action'); // 'approve' | 'reject'
$comments = post('comments');

if (!$reqId || !in_array($action, ['approve', 'reject'], true)) {
    setFlash('error', 'Invalid request.');
```

```
    redirect(BASE_URL . '/approvals/queue.php');
}

// Load the requisition
$req = fetchOne("SELECT * FROM requisitions WHERE requisition_id = ?", [$reqId]);

if (!$req) {
    setFlash('error', 'Requisition not found.');
```

```
    redirect(BASE_URL . '/approvals/queue.php');
}

// Use effective level
$effectiveLevel = getEffectiveApprovalLevel($user, $req);

// Make sure this req is actually at the user's effective level
```

```

if ((int)$req['current_approval_level'] !== $effectiveLevel) {
    setFlash('error', 'This requisition is not currently at your approval level.');
```

```

    redirect(BASE_URL . '/approvals/queue.php');
}

// Make sure it's still pending
if ($req['current_status'] !== 'pending') {
    setFlash('error', 'This requisition has already been ' . $req['current_status'] . '.');
    redirect(BASE_URL . '/approvals/queue.php');
}

// Reject requires a reason
if ($action === 'reject' && empty(trim($comments))) {
    setFlash('error', 'Please provide a reason for rejecting.');
```

```

    redirect(BASE_URL . '/approvals/queue.php?highlight=' . $reqId);
}

// Process the decision
processApprovalDecision($action, $reqId, (int)$user['user_id'], $comments);

// Set a friendly flash message
$reqNumber = $req['requisition_number'];
if ($action === 'approve') {
    $nextLevel = $effectiveLevel + 1;
    $msg = $nextLevel > 4
        ? "{$reqNumber} has been fully approved."
        : "{$reqNumber} approved and forwarded to " .
approvalLevelLabelForType($req['requisition_type'], $nextLevel) . ".";
} else {
    $msg = "{$reqNumber} has been rejected.";
}
setFlash('success', $msg);

$dest = ((int)($user['approval_level'] ?? 0) > 0)
    ? BASE_URL . '/approvals/queue.php'
    : BASE_URL . '/dashboard.php';
redirect($dest);

```

Appendix E: Manual Requisition Form



HR 01/02

ISUZU EAST AFRICA LIMITED PERSONNEL REQUISITION

DATE OF REQUEST 13/12/2024	NAME OF PERSON MAKING REQUEST Morris Ethare	TELEPHONE EXTENSION 342
REQUESTING DIVISION: Finance & Strategy		SECTION: IT
TITLE & BRIEF OF POSITION OR CLASSIFICATION TO BE FILLED: IT Atachee		DATE REQUIRED 13/01/2025
IDENTIFY THE ELEMENTS OF THE JOB - Assist in developing, testing and debugging software applications - Provide technical support to users, troubleshooting and resolving issues - Participate in IT training sessions to improve IT skills and knowledge - Observing and learning from experienced IT professionals		
SPECIAL QUALIFICATIONS DESIRED [LIST SKILLS, EXPERIENCE AND KNOWLEDGE] - Pursuing a degree in computer science, information technology, or a related field - Basic knowledge of programming, network systems, and Microsoft Office Suite - Good problem-solving, communication, and interpersonal skills		
ISUZU EAST AFRICA SPECIFIC TRAINING REQUIRED BEFORE ASSIGNMENT TO JOB [TYPE AND DURATION] e.g. SAP, Sage, Aurymaster, ILM, on-the job training etc. N/A		
WORKING TOOLS/ EQUIPMENT REQUIRED FOR THE JOB Desktop/Laptop		
REGULAR EMPLOYEE ☐	TEMPORARY ☒ 3 Months	IF TEMPORARY ESTIMATE TIME NEEDED
REPLACEMENT ☐	[SHOW NAME OF PREVIOUS INCUMBENT]	
ADDITION ☐	[GIVE REASON]	
APPROVALS		
RECOMMENDED BY DEPARTMENT HEAD	SIGNATURE:	DATE:
APPROVED BY DIRECTOR, HR & CORPORATE, SUSTAINABILITY DIVISION	SIGNATURE:	DATE:
APPROVED BY DIRECTOR FINANCE & STRATEGY DIVISION	SIGNATURE:	DATE:
APPROVED BY MANAGING DIRECTOR	SIGNATURE:	DATE:

* Positions to be filled internally - 1 month
 * Positions to be filled externally - 2 months

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